DX Atlas Pro

v.0.92 Beta

User's Guide



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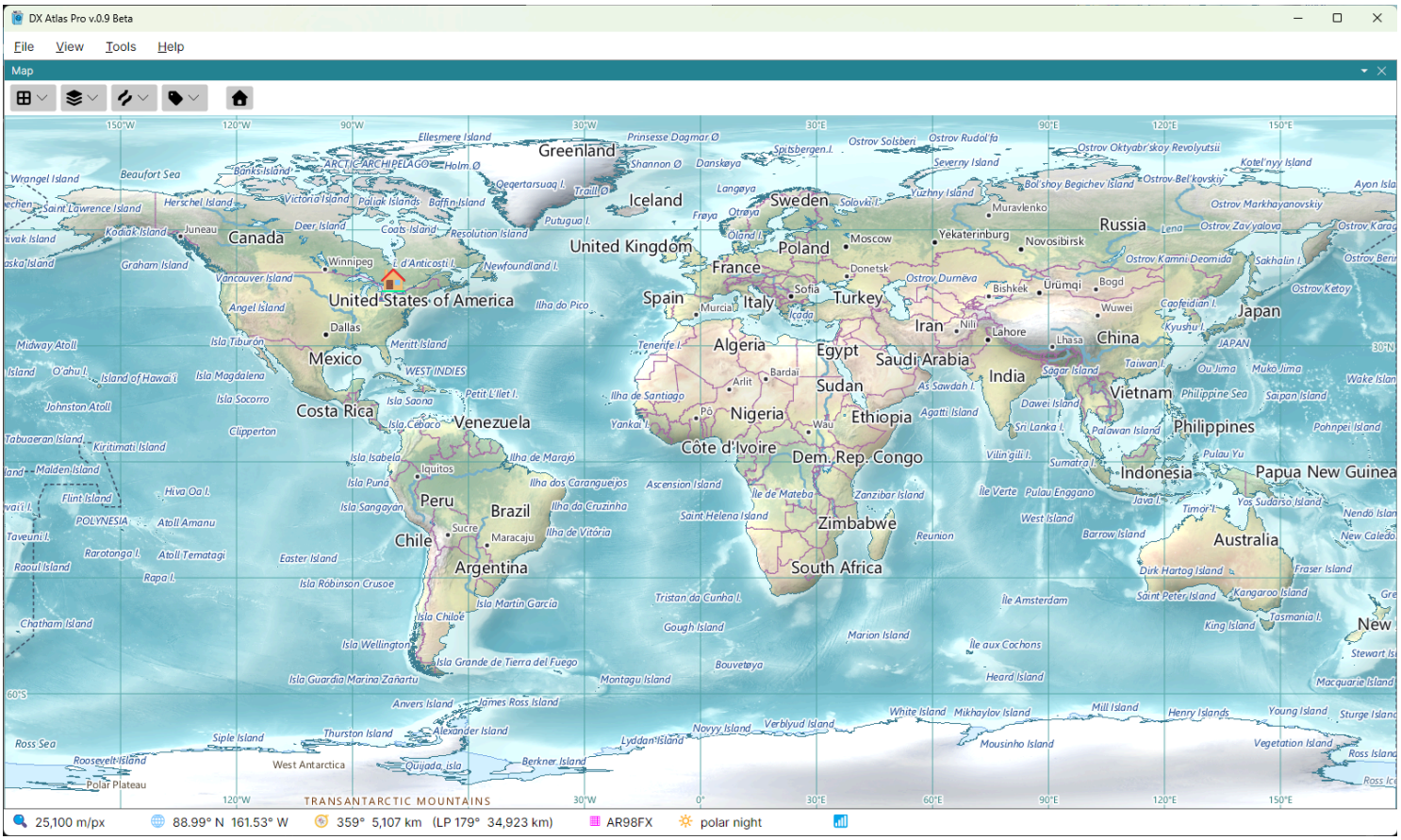
Overview



DX Atlas Pro is a world atlas for Windows and Linux. It draws the Earth with shaded relief, land cover and ocean depth, in three projections and on two independent map panels, with countries, cities, rivers, lakes and time zones, the day and night terminator, and a clock that can show the map at any date and time.

The program serves two audiences. With **Tools > Amateur Radio Functions** off, it is a general-purpose atlas, and none of the Ham Radio terms appear in its menus and lists. With the switch on, it becomes a tool for Radio Amateurs: DXCC entities and prefixes, prefix search, CQ and ITU zones, IOTA groups, grid squares, WAP Antarctic bases, award tracking, Great Circle paths, ionospheric maps, the auroral oval and HF propagation prediction.

DX Atlas Pro is a new program. It is not an upgrade of DX Atlas, and DX Atlas licences do not cover it; see Register.



Features

World Atlas

- three map projections: rectangular, azimuthal and globe, with zoom and pan;
- two map panels, each with its own projection, centre and zoom;
- shaded relief, land cover and ocean depth, with lakes, rivers, glaciers, ice shelves and urban areas;
- countries, provinces and states, and cities;
- time zones, and the local time of every country, city and island on the map, with summer time taken into account, at the current time or at any moment in the past since 1945;
- the day and night terminator, and sunrise and sunset at any point on the globe;
- the Clock panel: UTC or local time;
- the Lists panel: the items of every layer in lists, with search, a click on a row to find the item on the map and a click on the map to find it in the list;
- item details: the countries, zones, islands and other items at a point on the map, with their attributes;
- pins, and layers of your own, loaded from JSON files;

- named views, saved and restored from the menu;
- panels that can be docked or floated, light and dark themes, and the map colours set by the user;
- an automation API for other programs: a WebSocket API, and a COM interface compatible with DX Atlas 2 on Windows; see the Developer's Guide.

Amateur Radio

- prefix search: the DXCC entity of a callsign, found from its prefix;
- DXCC entities and their prefixes, in a tree of entities, their call areas and provinces;
- CQ zones, and ITU zones in both the ARRL and the RSGB versions;
- IOTA groups;
- VUCC grid squares and FFMA squares;
- WAP Antarctic bases;
- award tracking: lists of the entities you worked or confirmed, per band or mode, shown in colour on the map;
- Great Circle paths from your home location, short path and long path;
- ionospheric maps computed with the IRI-2026 model: F2 layer critical frequency (foF2), MUF for 3000 km, F2 layer peak height (hmF2), E layer critical frequency (foE), D layer peak electron density (NmD) and magnetic dip. The maps reflect the current conditions using the sunspot number and the Ap index downloaded automatically;
- the auroral oval in real time, from the NOAA Space Weather Prediction Center;
- HF propagation prediction maps computed with the DVOACAP engine: the signal strength from your station on the selected band and hour, updated in real time to reflect geomagnetic disturbances using the STORM model, and maps pre-computed for all bands and hours of the day.

The real-time data - the auroral oval, the Ap index and the sunspot numbers - are available only as long as the NOAA Space Weather Prediction Center, the GFZ Helmholtz Centre for Geosciences and irimodel.org provide them on the web.

System Requirements

Hardware

- **Computer**: 64-bit PC with an x64 processor;
- **Video Card**: on Windows, any video card with a Direct3D 11 driver; on Linux, OpenGL 3.3 or higher;
- **Disk Space**: about 200 MB;
- **Internet**: optional. Without it, the program works fully as an atlas; the connection is used to download the sunspot number and the Ap index for the ionospheric and propagation maps, and the auroral oval in real time.

Software

- **Windows**: Windows 10 or Windows 11, 64-bit. The setup program installs the .NET 10 Runtime and the Microsoft Visual C++ Redistributable if they are not on the computer yet;

- **Linux**: 64-bit x64 Linux with an X11 desktop and an OpenGL driver. The program is self-contained and needs no .NET Runtime. Tested on Ubuntu 24.04 and Linux Mint 22.3. The OpenGL driver is normally installed with the desktop.

  To see which display system your desktop uses, type `echo $XDG_SESSION_TYPE` in a terminal: it prints `x11` or `wayland`. On a Wayland desktop the program runs through XWayland, which most distributions install with the desktop; this has not been tested yet, so if the program does not start, choose an X11 (often called "Xorg") session on the login screen.

Installation

Download the setup program for Windows, or the archive for Linux, from the Download page. Check the System Requirements first.

Windows

1. Run `DxAtlasProSetup-v.X.Y.exe`, where X.Y is the version number. The setup needs administrator rights, and Windows asks for them when it starts;
2. accept the Licence Agreement;
3. if the .NET 10 Runtime or the Microsoft Visual C++ Redistributable is not on the computer, the setup downloads and installs it;
4. optionally, tick **Create a desktop shortcut**;
5. when the setup finishes, leave **Launch DX Atlas Pro** ticked to start the program.

The program is installed in `C:\Program Files\Afreet\DxAtlasPro`, with a shortcut in the Start menu. The setup also registers the COM interface that lets programs written for DX Atlas 2 work with DX Atlas Pro; see the Developer's Guide.

To install a new version, run its setup over the old one. Your settings are kept.

Uninstalling

Uninstall DX Atlas Pro from **Settings > Apps > Installed apps** in Windows. Your settings are not removed; they stay in `%APPDATA%\Afreet\Products\DxAtlasPro`, and you can delete that folder by hand.

If DX Atlas is also installed on the computer, uninstalling DX Atlas Pro removes the COM registration that the two programs share, so reinstall DX Atlas afterwards for the programs that use its COM interface.

Linux

The Linux version is a self-contained archive: it needs no installation and no .NET Runtime.

1. Unpack the archive to any folder, for example your home folder; it creates a `DxAtlasPro` folder there, so do not create one yourself:

```
tar -xzf ~/Downloads/DxAtlasPro-v.X.Y-linux-x64.tar.gz -C ~
```

Replace X.Y with the version number, and `~/Downloads` with the folder you downloaded the archive into if your browser saved it elsewhere;

2. run the program:

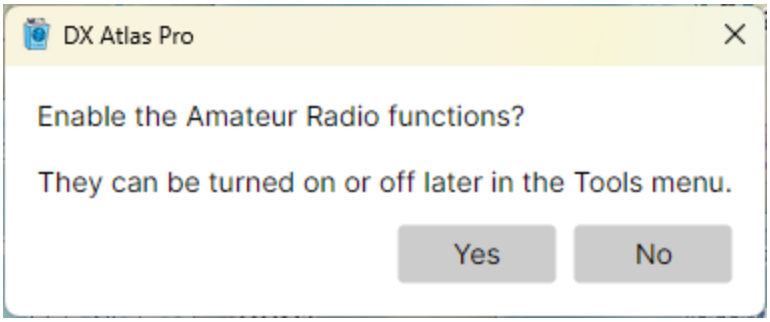
```
~/DxAtlasPro/DxAtlasPro
```

To install a new version, unpack it over the old one. Your settings are kept in `~/.config/Afreet/Products/DxAtlasPro`. To uninstall, delete the program folder and, if you do not need your settings any more, that folder too.

Quick Start

First Start

When DX Atlas Pro starts for the first time, it asks whether to enable the Amateur Radio functions:



- answer **Yes** to show the DXCC entities, zones, award tracking, ionospheric maps and propagation prediction;
- answer **No** to use the program as a general-purpose world atlas.

The answer is not final: **Tools > Amateur Radio Functions** turns the functions on and off at any time. See The Amateur Radio Switch.

Setting the Home Location

The home location is your own location on the globe. It is marked on both maps, the distances and bearings in the status bar are measured from it, the Clock panel shows its sunrise and sunset, and the azimuthal map is centred on it.

To set it, do one of the following:

- right-click your location on the map and pick **Set Home Location Here**, then confirm the coordinates in the message box;
- open **Tools > Settings**, and enter your latitude and longitude, or your Maidenhead grid square, in the **Home Location** section.

See Home Location.

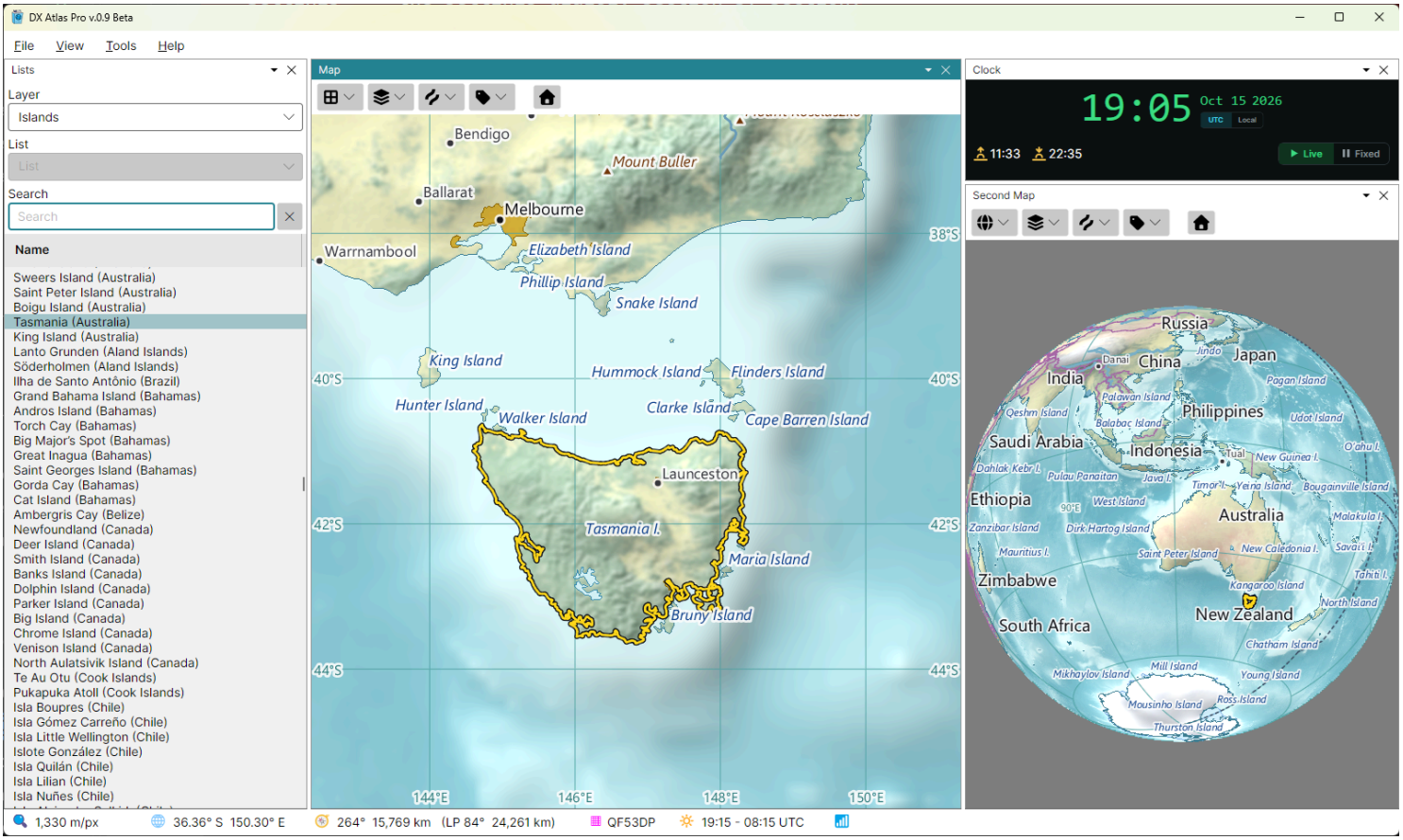
Finding Your Way Around

- drag the map with the left mouse button to pan it, and turn the mouse wheel to zoom in and out;

- click the **Projection** button at the left end of the map toolbar to switch between the rectangular map, the azimuthal map and the globe;
- click the **Layers**, **Lines** and **Labels** buttons to choose what the map shows;
- hover the mouse over a label to see its tooltip, and right-click the map for a menu of what you can do at that point;
- type a name in the **Search** box of the **Lists** panel, then double-click a row to find that item on the map.

The rest of this guide describes each of these in detail. Window Layout explains the panels of the main window.

Window Layout



The main window holds the main menu at the top, the status bar at the bottom, and four panels between them:

- **Lists** - the layers of the map as lists of items, with search; see The Lists Panel;
- **Map** and **Second Map** - two independent map panels, each with its own toolbar, projection, centre, zoom and layers; see The Map;
- **Clock** - the current time, or a time of your choice for the gray line; see Clock and Fixed Time.

Showing and Hiding Panels

The **View** menu has a checkbox for each panel: tick it to show the panel, clear it to hide the panel. A panel can also be closed by the close button on its tab. When the program starts for the first time, the Lists panel and the Map are shown, and the Second Map and the Clock panel are hidden; show them with **View > Second Map** and **View > Clock**.

Closing a panel has these side effects:

- closing the **Clock** panel returns the clock to the computer's time, so the maps do not stay at a forgotten fixed time;
- closing the **Lists** panel removes the highlight, the pins and the award colouring from the maps, and showing it again puts them back.

The status bar is always shown.

Docking and Floating

Each panel can be docked - placed inside the main window, beside, above or below another panel, or on a tab in the same group as another panel - or floated in a window of its own.

Moving a Panel

To move a docked panel, drag its tab, the strip with its name at the top of the panel. To move a floating panel, drag the title bar of its window.

While you drag, docking targets appear over the panel under the mouse cursor: a cross of five buttons in the middle of that panel, and a button at the edge of the main window. Each button shows, in its picture, where the panel will go:

- the **top**, **bottom**, **left** and **right** buttons of the cross split the panel under the cursor, and put the dragged panel above it, below it, or beside it;
- the **centre** button puts the dragged panel on a tab in the same group as the panel under the cursor, so that the two take turns in the same place;
- a button at the **edge of the main window** docks the dragged panel along that whole edge.

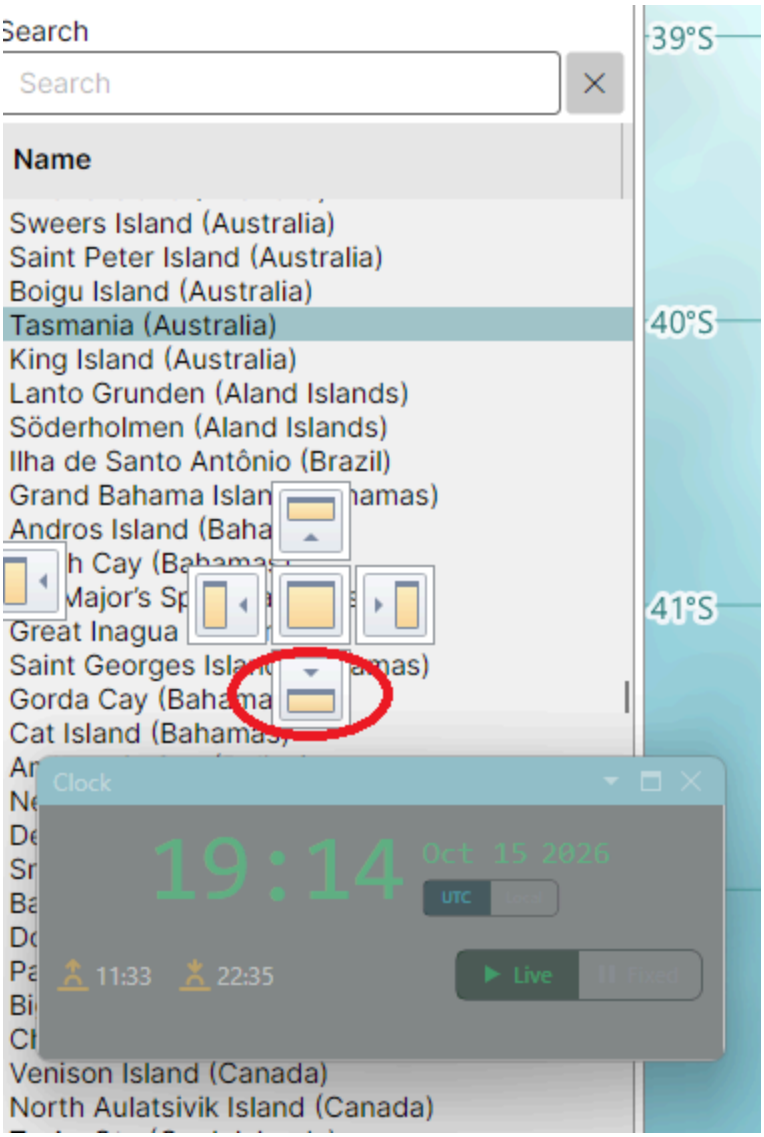
What counts is where the **mouse cursor** is when you release the button, not where the dragged panel or its window is: move the cursor itself onto the target button. Releasing it anywhere else leaves the panel floating in a window of its own, which you can move to another monitor.

Example: the Clock Below the Lists

To dock the Clock panel below the Lists panel:

1. show the Clock panel with **View > Clock**;
2. drag the Clock panel by its tab, or by the title bar of its window if it is floating, over the Lists panel. The docking targets appear over the Lists panel;

3. move the mouse cursor onto the **bottom** button of the cross, the one circled in the picture below, and release the mouse button.



The Lists panel is split in two, the list above and the Clock below.

Sizes and Saving

Drag the splitters between the panels to resize them. The layout, including the size and position of any floating window, is saved when the program closes and restored the next time it starts. To save several layouts and switch between them, use Named Views.

Themes

Tools > Theme switches the program between the **Light** and the **Dark** theme. The theme changes the menus, panels and dialogs; the map colours are set separately, in the Colors Dialog. The Clock panel is dark in either theme.

Projections, Zoom and Pan

Each of the two map panels has a toolbar above the map:



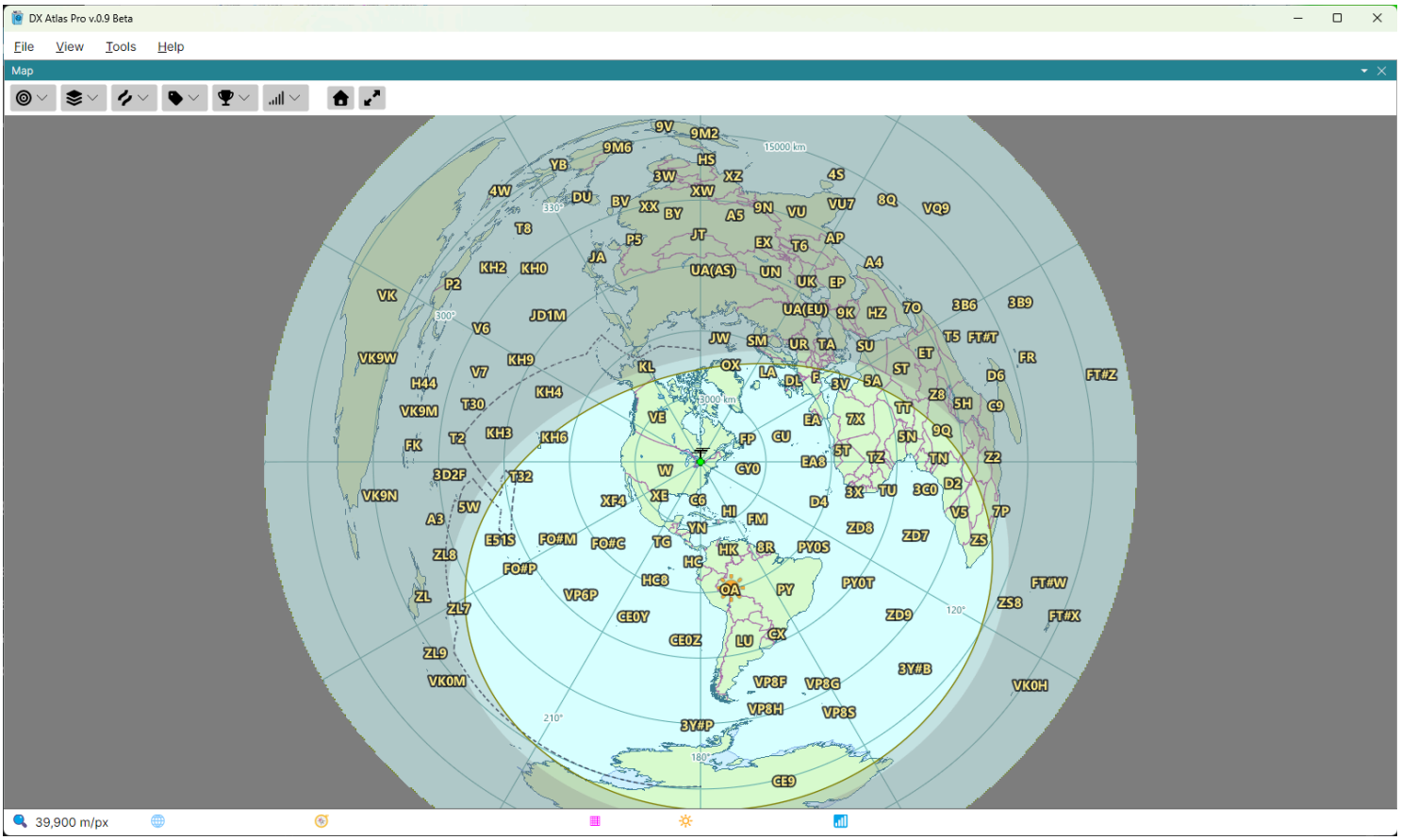
- **Projection** - the projection of this map;
- **Layers**, **Lines** and **Labels** - what the map shows; see Layers, Lines and Labels;
- **Amateur Radio** and **Ionosphere** - shown only with the Amateur Radio functions on; see The Amateur Radio Switch;
- **Home** - see Home Location.

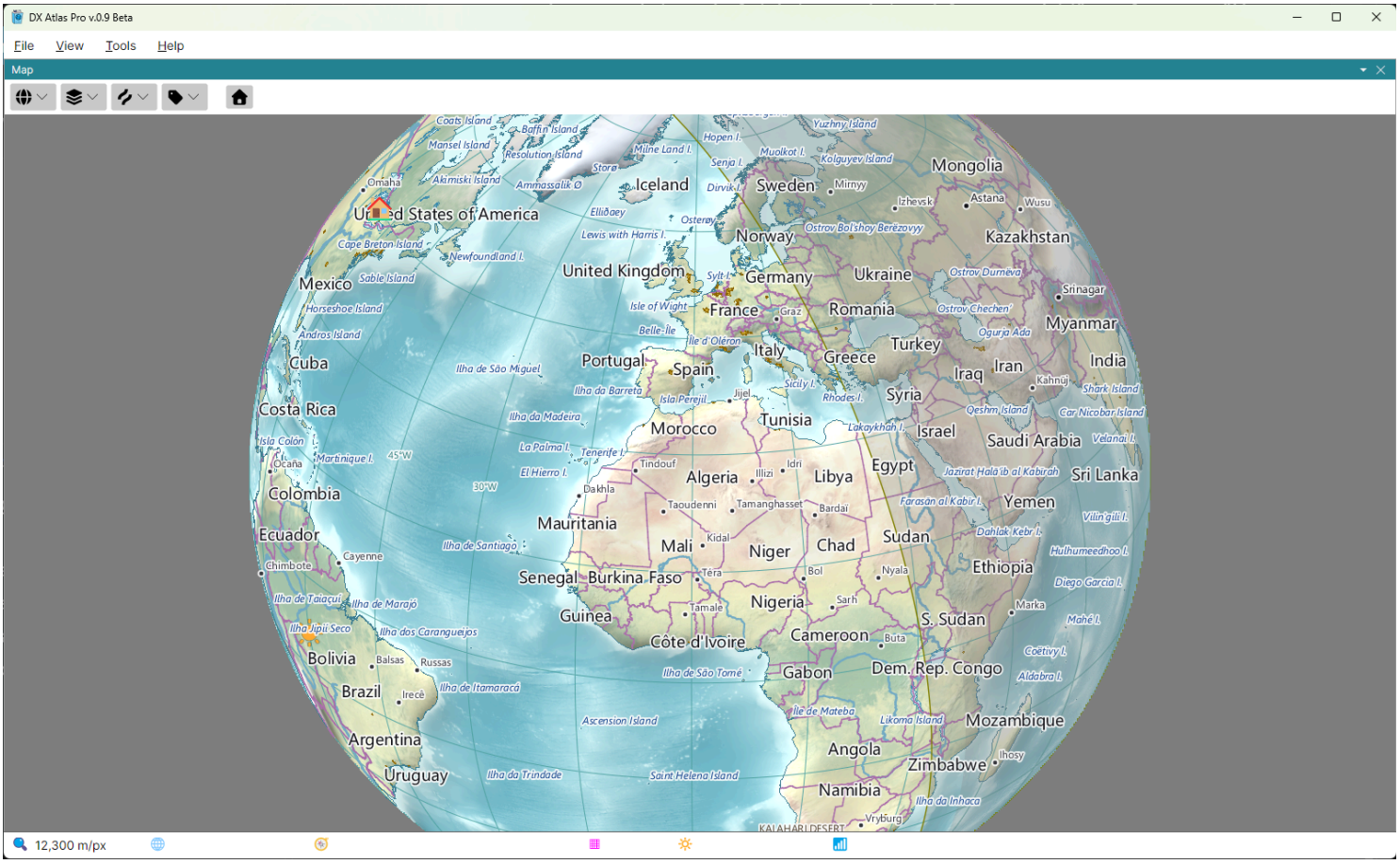
The two maps are independent: each has its own projection, centre, zoom and layers.

Projections

Click the **Projection** button and pick one of the three projections. The button shows the glyph of the current one.

- **Rectangular** - the whole world in one rectangle, latitude and longitude as straight lines;
- **Azimuthal** - the azimuthal equidistant projection, centred on one point: every straight line through the centre is a great circle, and the distance and the bearing from the centre to any point are true. When you switch to it, the map is centred on your home location;
- **Globe** - the Earth as seen from space.


DX Atlas Pro v.0.9 Beta
File View Tools Help
Map
39,900 m/px


DX Atlas Pro v.0.9 Beta
File View Tools Help
Map
12,300 m/px

Zoom

Turn the mouse wheel over the map to zoom in and out; three steps of the wheel double or halve the magnification. On the rectangular map the zoom keeps the point under the mouse in place, so the wheel takes you to a place as well as closer to it. The azimuthal map and the globe zoom about their centre, which stays where it is.

The status bar shows the scale of the map under the mouse, in metres or feet per pixel, as set in the **Units** section of the Settings Dialog.

Pan

- **Rectangular map** - drag the map with the left mouse button;
- **Globe** - drag the globe to turn it;
- **Azimuthal map** - its centre is chosen, not dragged, since every distance and bearing on it is measured from the centre. Shift-click a point to make it the centre, or right-click it and pick **Set as Center**. Ctrl+Shift-click a point, or pick **Set Antipode as Center**, to centre the map on the point opposite it on the globe.

The **Home** button on the toolbar shows the whole world on the rectangular map, and centres the azimuthal map and the globe on your home location.

Linking the Two Maps

Alt-click a point on one map, or right-click it and pick **Center Other Map Here**, to centre the other map on that point; the other map keeps its projection and zoom. This works while both maps are shown.

The keys held with each click can be changed in the **Mouse Clicks** section of the Settings Dialog.

The Map's Context Menu

Right-click the map for a menu of what can be done at that point:

- **Show Details for ...** - the Item Details of the label under the mouse, or of everything at the point;
- **Find in List** - select the item under the mouse in the Lists Panel;
- **Set as Center** and **Set Antipode as Center** - on the azimuthal map only;
- **Center Other Map Here**;

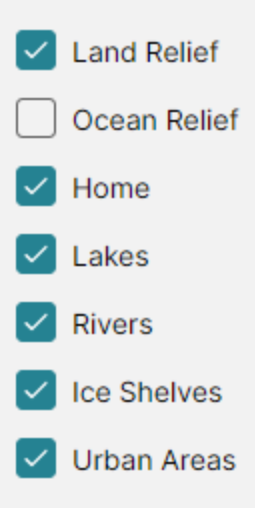
- **Set Home Location Here** - see Home Location;
- **Add Pin** and **Remove Pin** - see Pins.

Layers, Lines and Labels

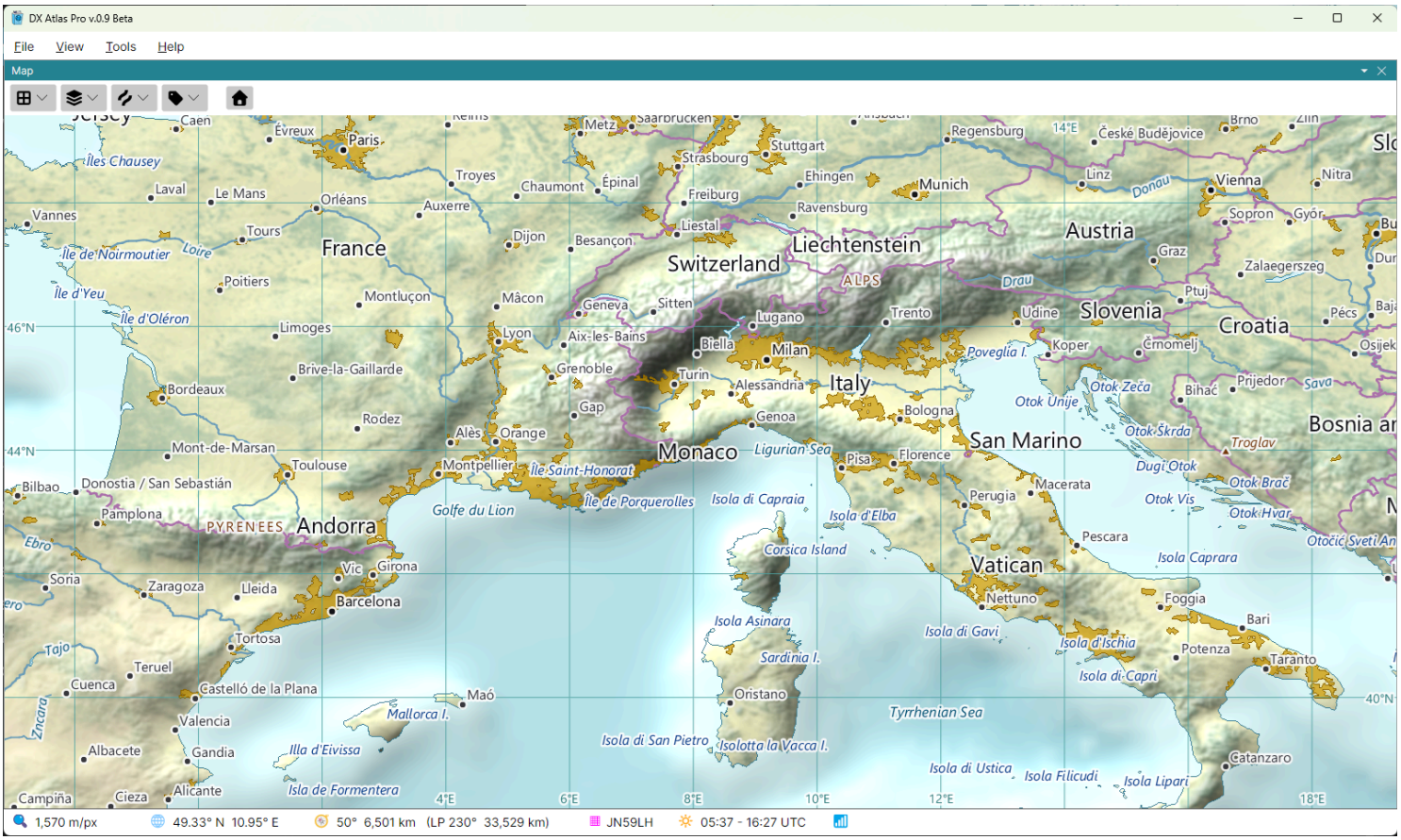
Three buttons on the map toolbar choose what the map shows: **Layers**, **Lines** and **Labels**. Each opens a list of checkboxes; tick a checkbox to show its layer, clear it to hide the layer. The choices are made for each map separately, and saved when the program closes.

The Amateur Radio layers - DXCC entities, zones, IOTA groups and the rest - are in a fourth button's menu, shown only with the Amateur Radio functions on; see The Amateur Radio Switch.

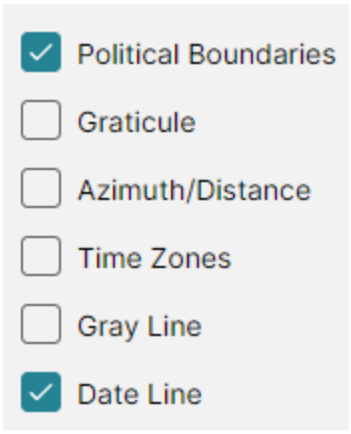
Layers



- **Land Relief** - the shaded relief and land cover of the land. Cleared, the land is one flat colour;
- **Ocean Relief** - the depth and the shaded relief of the sea floor. Cleared, the sea is one flat colour. Clear both to read boundary lines against a plain map; the two flat colours are set in the Colors Dialog;
- **Home** - the mark of your home location, a house, or an antenna with the Amateur Radio functions on;
- **Lakes**, **Rivers**, **Ice Shelves** and **Urban Areas**.



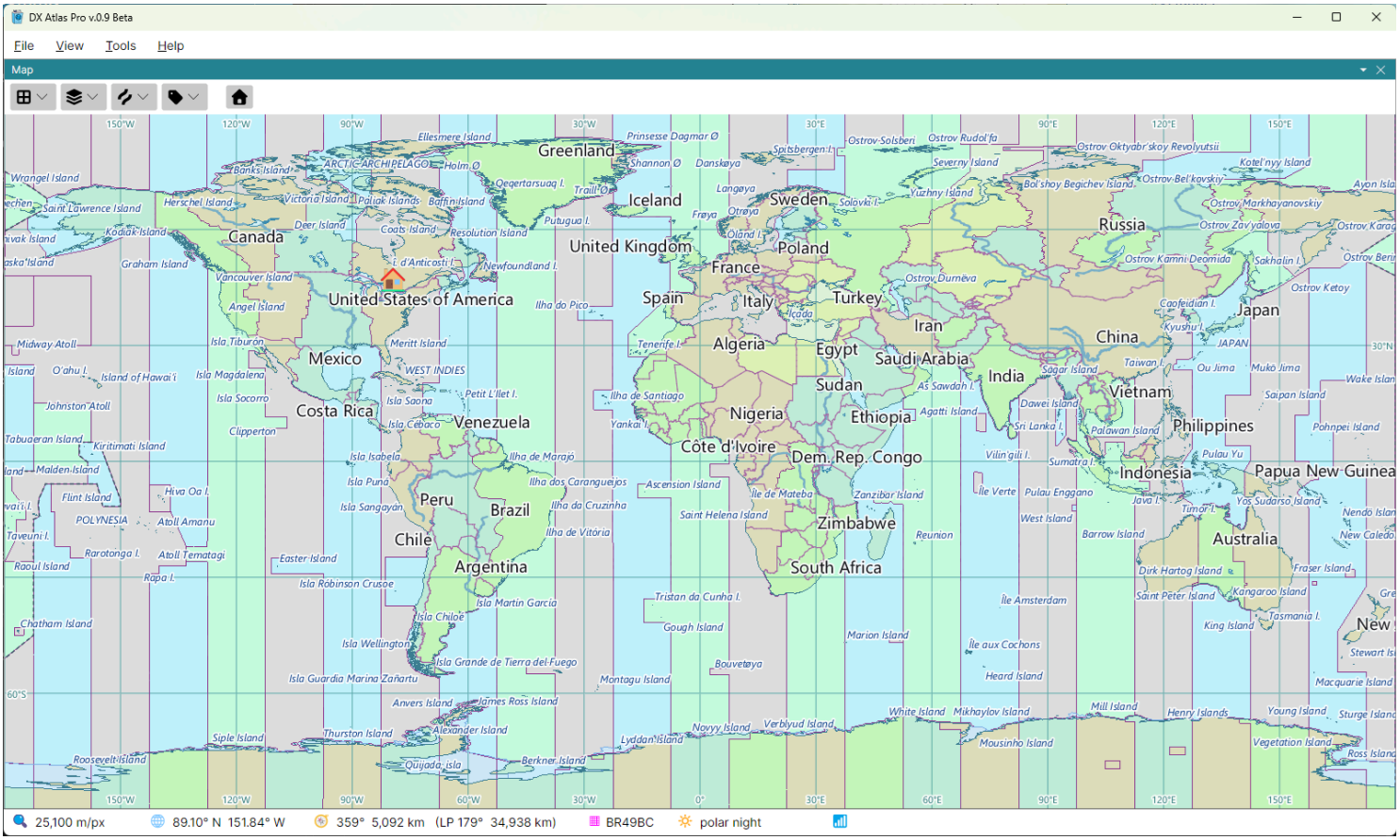
Lines

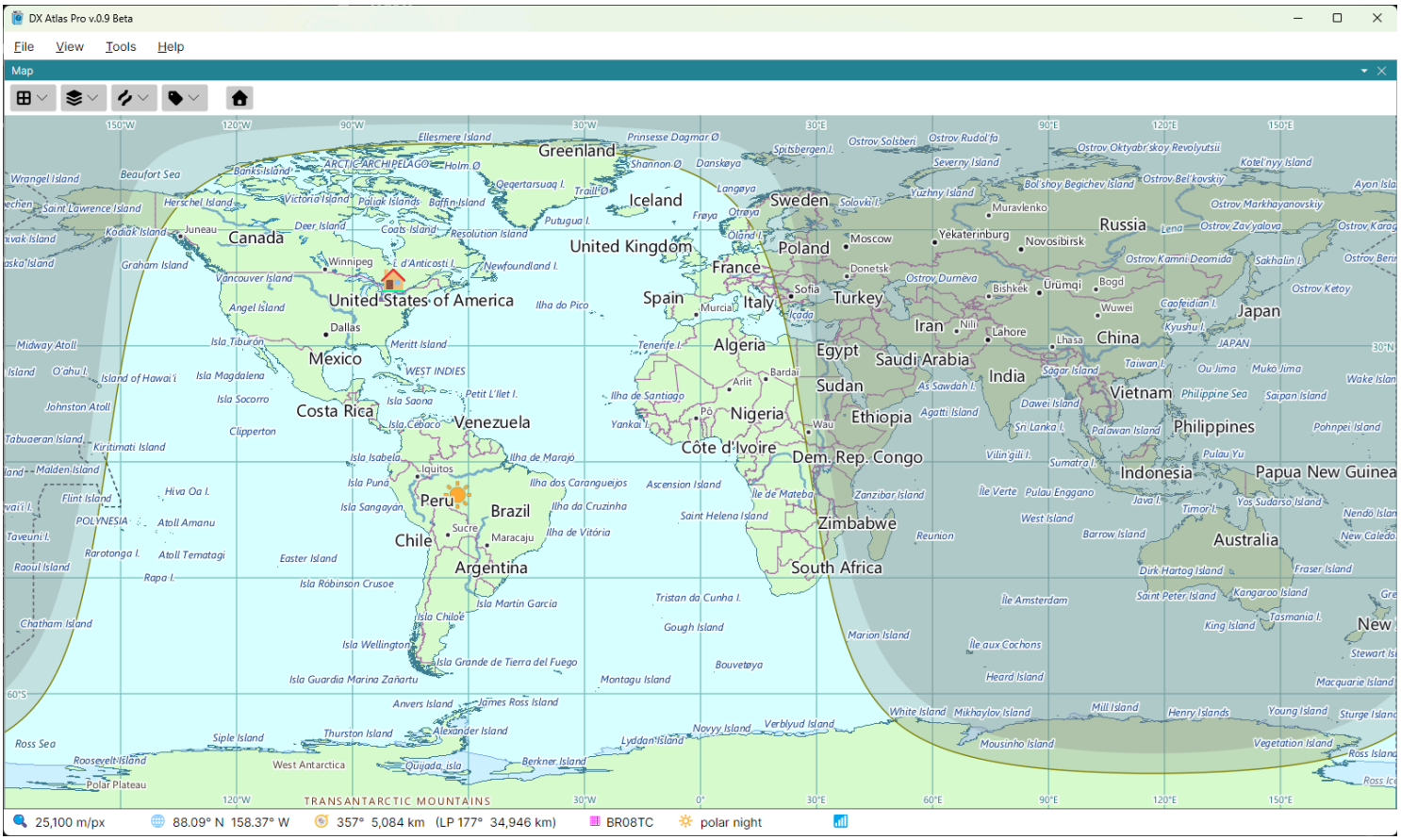


- **Political Boundaries** - the borders between countries;
- **Graticule** - the lines of latitude and longitude, labelled with their degrees;
- **Azimuth/Distance** - great circles radiating from your home location, and circles of equal distance around it, labelled with their bearings and distances. The distance circles are drawn every 3000 km, the length of an F2 hop. On the azimuthal map they radiate from the centre of the map;
- **Time Zones** - the time zones, filled in alternating colours;
- **Gray Line** - the day and night terminator and the night shadow, with the sun at the point where it is overhead;

- **Date Line** - the international date line.

The coastline has no checkbox: it is always drawn.

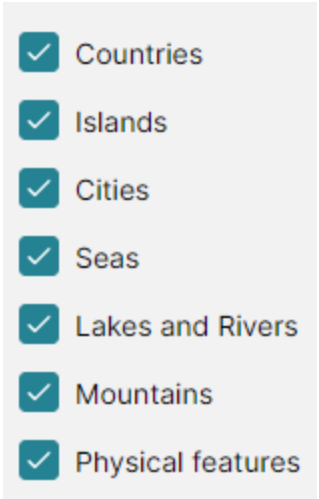




The Gray Line

The gray line follows the sun as the time goes by, at the time the Clock panel shows: the computer's time, or a fixed time of your choice. Which of its parts are drawn - the sun, the terminator, the twilight shadow and the night shadow - where the night shadow begins, and how often it is redrawn are set in the **Map** section of the Settings Dialog.

Labels



- **Countries**;
- **Islands** - islands and island groups;
- **Cities**;
- **Seas** - oceans, seas, gulfs, bays and ice shelves;
- **Lakes and Rivers**;
- **Mountains** - mountain ranges and summits;
- **Physical features** - deserts, plateaus, peninsulas, capes and other regions.

The labels are placed so that they never overlap. At a world view only the largest features are labelled, and more labels appear as you zoom in.

Tooltips

Hover the mouse over a label to see its tooltip: the full name of the item, or, for the Amateur Radio labels, the name of a DXCC entity or province, or the definition of a zone or a call area. Click a label to see the Item Details of the item it names.

Colours

The colours and line widths of all layers, and the colours, sizes and fonts of all labels, are set in the Colors Dialog.

Home Location

The Home Location

The home location is your own location. It is used in several places:

- both maps mark it with a house. The mark is switched on and off by **Home** in the Layers menu;
- the status bar shows the bearing and the distance from it to the point under the mouse;
- the azimuth and distance lines radiate from it;
- the azimuthal map is centred on it when you switch to that projection, and again whenever the home location changes;
- the Clock panel shows its sunrise and sunset, and can lock the time to them.

There are three ways to set it:

- right-click the map at your location and pick **Set Home Location Here**. A message box shows the latitude, longitude and grid square of the point; click **Yes** to make it home;
- enter the latitude and longitude in the **Home Location** section of the Settings Dialog;
- enter your 4- or 6-character Maidenhead grid square in the same section; the latitude and longitude move to the centre of that square.

A click with keys of your choice can also set the home location: assign them to **Set as Home** in the **Mouse Clicks** section of the Settings dialog.

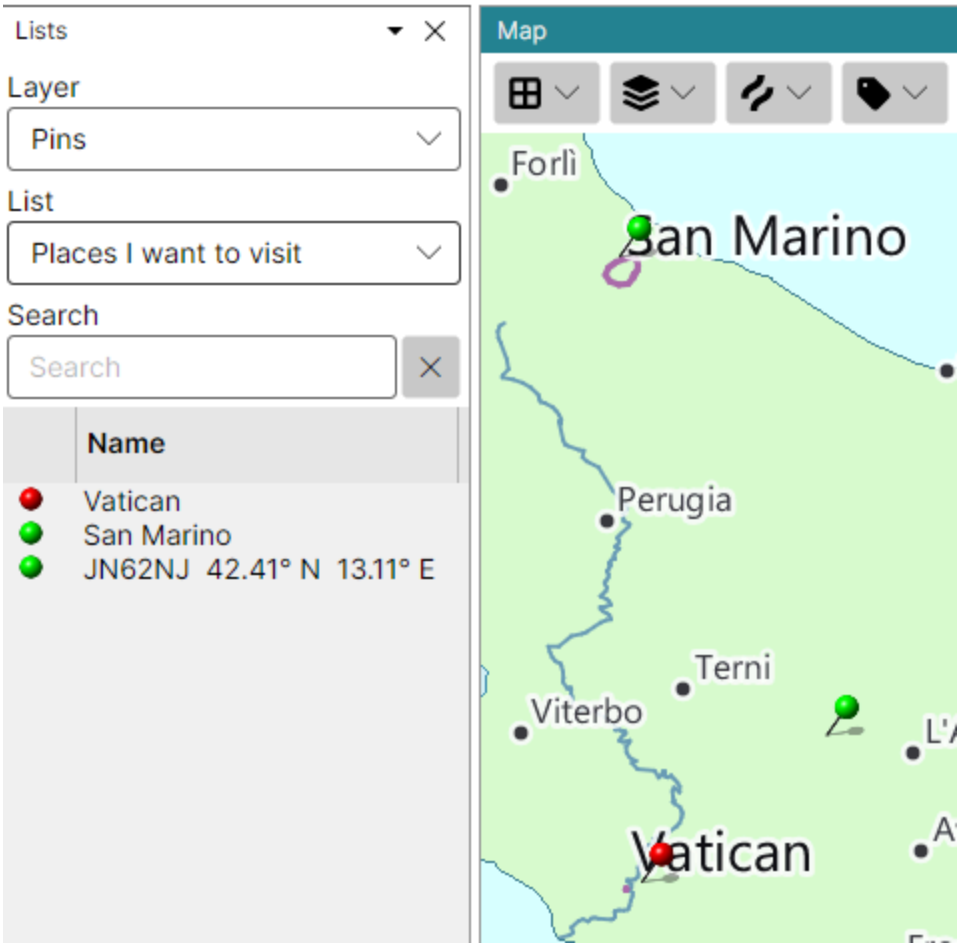
The Home Button

The **Home** button on the map toolbar zooms the map out to the whole world:

- the rectangular map is centred on latitude and longitude 0;
- the azimuthal map and the globe are centred on the home location.

Pins

Pins mark places of your choice on both maps: a friend's location, a place you plan to visit, a station you worked. They are kept in lists on the **Pins** layer of the Lists Panel, and the maps show the pins of the list selected there.



Creating a Pin List

Pins can be added only once a pin list exists:

1. in the Lists panel, pick **Pins** in the **Layer** box;
2. in the **List** box, pick **New List...** and type a name for the list.

You can keep several pin lists - one per trip, say - and switch between them in the List box. The maps show the pins of the list selected on the Pins layer, even while the Lists panel shows another layer.

Adding a Pin

Right-click the map where the pin should go, open **Add Pin**, and pick one of the ten colours. Without a pin list the item is disabled, and its tooltip says why.

A new pin is named by its grid square and its latitude and longitude.

Editing and Removing Pins

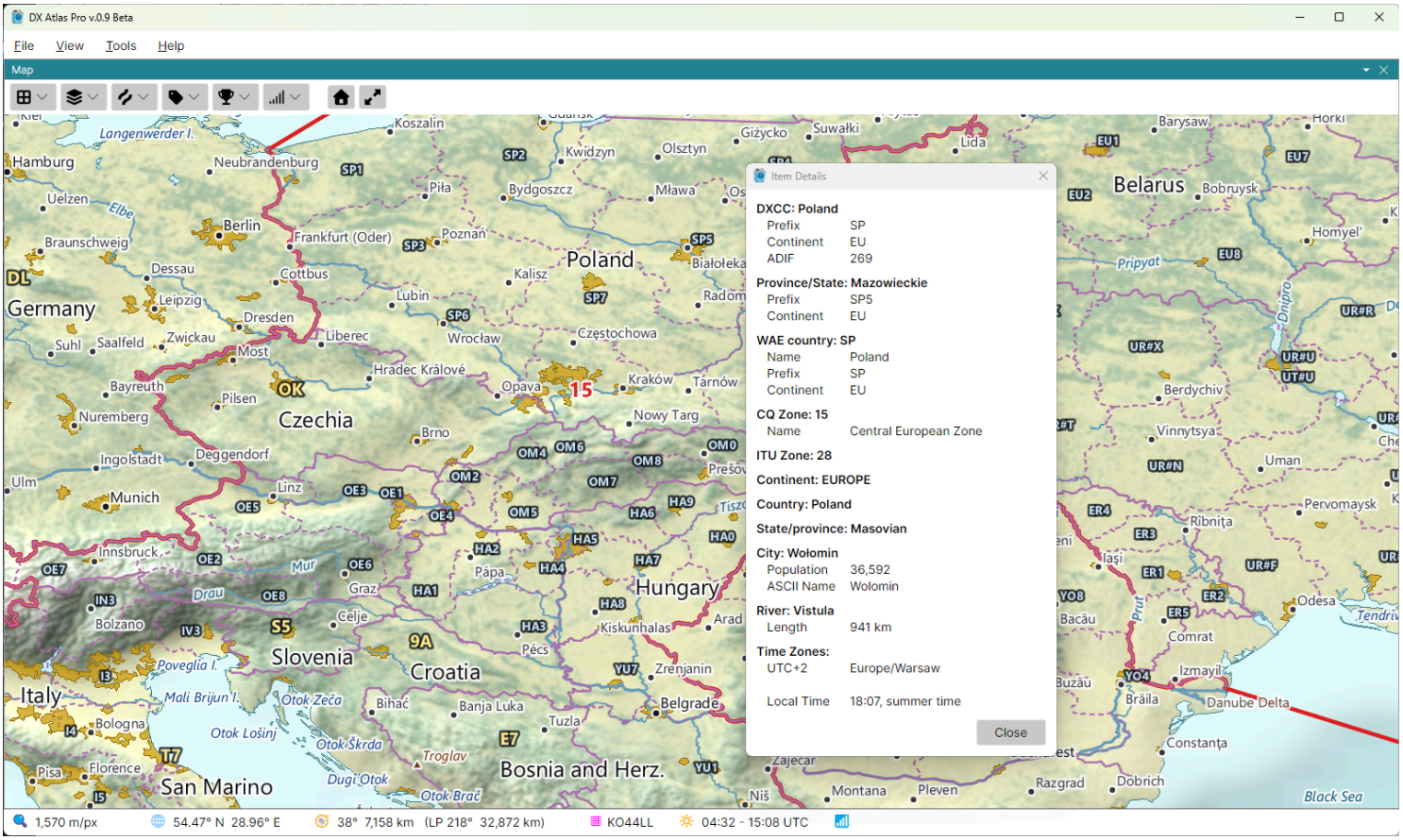
- **On the map** - right-click a pin and pick **Remove Pin**;
- **In the Lists panel** - with the Pins layer selected, right-click the row of a pin:
  - **Edit Pin...** opens a dialog in which you can give the pin a name of your own and another colour. An empty name names the pin by its grid square and position again;
  - **Remove Pin** removes it.

Double-click a row to find its pin on the map. The pins are saved as soon as they change.

Pins are shown only while the Lists panel is open; closing the panel takes them off the maps.

Item Details

The **Item Details** window lists what is at a point of the map: the country, the state or province, the city, island, sea or lake, and, with the Amateur Radio functions on, the DXCC entity, its call area and province, and the zones, each with its attributes. The time zones and the local time at the point come last.



Opening the Window

- **Ctrl-click the map** to list everything at the point clicked;
- **click a label** to list the item the label names, as if you clicked its place on the map;
- **right-click the map** and pick **Show Details for ...**: the menu item names the label under the mouse, or says **Point** where there is no label;
- in the Lists Panel, **Ctrl-click a row**, or right-click it and pick **Show Details**, to list that row's item. A province or call area is listed with its DXCC entity above it.

The keys of the Ctrl-click can be changed in the **Mouse Clicks** section of the Settings Dialog.

What the Window Shows

Each item has a heading - its type and name, such as **Country: Japan** - and its attributes indented under it. The attributes depend on the item: an island lists its area, a city its population, a DXCC entity its prefix, zones, continent and ADIF number. A detail that another item already gives is not repeated: the zones are listed once, under their own headings.

The local time is given for the moment the Clock panel shows, with summer time taken into account, so with a fixed time on the clock you can see the local time anywhere at any moment since 1945.

The position, grid square, bearing, distance and sunrise and sunset of the point are not listed: the Status Bar shows them as you move the mouse.

Closing the Window

The window stays open, and each new click fills it again. To close it, click its **Close** button, press **Escape**, or click anywhere in the main window; that click does nothing else.

Named Views

A named view saves the state of the whole program under a name, so you can return to it with one click: a "Europe" view for one task and a "World, two maps" view for another. A view holds:

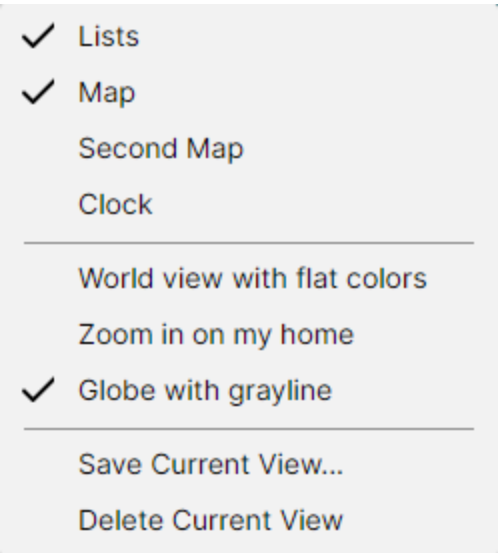
- whether the Amateur Radio functions are on;
- which panels are shown, and the window layout;
- for each map shown: its projection, its centre and zoom, and every checkbox of its Layers, Lines, Labels, Amateur Radio and Ionosphere menus.

Saving a View

Set the program up as you want it, then pick **View > Save Current View...** and type a name. Each view must have a name of its own.

Switching Views

The saved views are listed in the **View** menu, below the panel checkboxes. Pick one to restore it. The view you picked last is ticked in the menu for as long as the program still shows it unchanged; a pan, a zoom or a changed checkbox clears the tick. The sizes and positions of the panels are not compared.



Deleting a View

Pick the view to make it current, then **View > Delete Current View**. The item is disabled while no view is ticked.

Clock and Fixed Time

The **Clock** panel shows the time and the date, and the sunrise and sunset at your home location. It also sets the time the maps are drawn for: the computer's time, or a fixed time of your choice. The panel is hidden when the program starts for the first time; show it with **View > Clock**.

UTC and Local Time

The **UTC** and **Local** switch selects the time zone the panel shows: UTC, or the time zone of the computer. The sunrise and sunset in the Status Bar follow the same switch.

Live and Fixed Time

The **Live** and **Fixed** switch selects the time:

- **Live** - the panel follows the computer's clock, and the time is shown in green;
- **Fixed** - the clock stops at the current time, which is shown in amber, and the panel shows the buttons to change it.



While the time is fixed, everything that depends on the time follows the panel instead of the computer's clock: the gray line, the local times in the Item Details, the sunrise and sunset in the Status Bar, and, with the Amateur Radio functions on, the ionospheric maps and the aurora. You can see the gray line of any day of the year, or the local time anywhere at any moment.

Changing a Fixed Time

- the time buttons step the time by **1 hour**, **15 minutes** or **1 minute**, back and forward;

- the date buttons step the date by **1 month**, **1 week** or **1 day**, back and forward;
- right-click the time or the date and pick **Set Date...** to pick a day in a calendar. The time of day is kept. Picking a date while the clock is live fixes it.

Sunrise and Sunset Locks

The two buttons at the ends of the time row lock the time to the day's sunrise or sunset at your home location. While a lock is on, stepping the date keeps the time at sunrise or sunset of the new day, so you can follow the gray line at your sunrise through the seasons. Click the button again to release the lock.

Returning to the Computer's Time

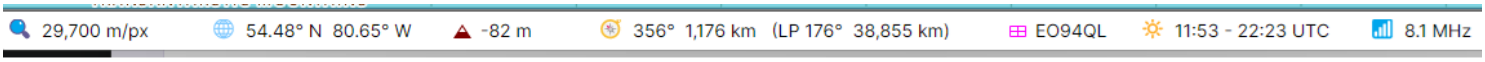
Click **Live**. Closing the Clock panel also returns to the computer's time, so the maps never stay at a fixed time you cannot see.

The Panel's Shape

The panel arranges itself to the room it has: a box, a narrow column, or, docked wide and short across the window, a strip. Its digits grow and shrink with it.

Status Bar

The status bar at the bottom of the main window describes the point under the mouse, on either map:

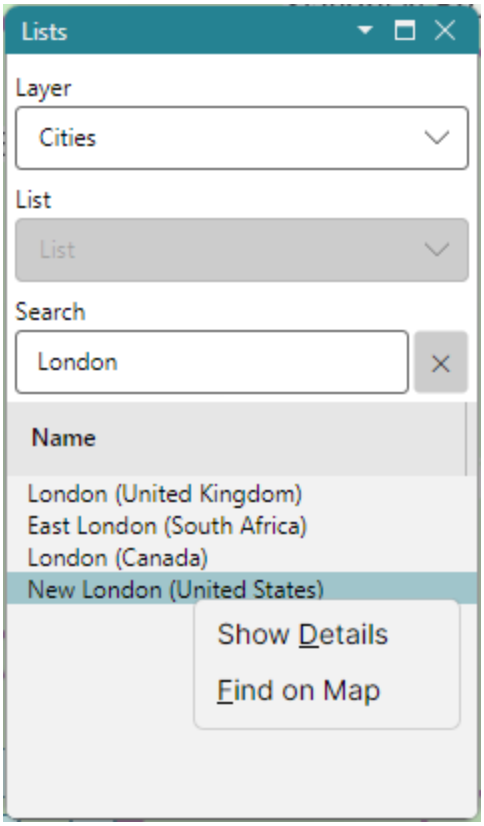


- 🔍 **Map scale** - the scale of the map under the mouse, in metres or feet per pixel. While the mouse is off the maps, the scale of the map you last clicked;
- 🌐 **Latitude and longitude** of the point;
- ▲ **Elevation** - the height of the ground or the ice above sea level at the point, or the depth of the sea there as a negative number. It is an average over about 15 km, so a summit reads lower than its true height, and a point near the coast reads between the land and the sea;
- 🧭 **Azimuth and distance from home** - the bearing and the distance from your home location to the point along the great circle, then, after **LP**, the same along the long path: the opposite bearing and the rest of the great circle around the Earth. While you drag the mouse with the Great Circle button down, they are measured from the point where you pressed the mouse button instead of from home;
- ⊞ **Grid square** - the 6-character Maidenhead locator of the point;
- 🔆 **Sunrise and sunset** at the point, in UTC or local time as the **UTC** and **Local** switch of the Clock panel selects, on the day the panel shows, or **polar day** or **polar night** where the sun does not rise or set that day;
- 📶 **Map reading** - with the Amateur Radio functions on, the value of the ionospheric or propagation map at the point; see Ionospheric Maps.

Hover the mouse over a value to see what it is. The distances, the elevation and the map scale are shown in the units chosen in the **Units** section of the Settings Dialog: kilometres and metres, miles and feet, or nautical miles and feet.

The Lists Panel

The **Lists** panel shows the items of a map layer - countries, cities, islands, rivers and the rest - as a list that you can search and sort, and links that list to the maps: pick a row to find its item on the map, or pick an item on the map to find its row.



The panel has three boxes at the top and the list of items under them.

Layer

The **Layer** box selects the layer the list shows:

- the layers of the atlas, named and grouped as in the **Labels** menu of the map, so that every name on the map is a row of one of them, and every row a name on the map:
    - **Countries**;
    - **Islands** - islands and island groups;
    - **Cities**;
    - **Seas** - oceans, seas, gulfs, bays, straits and ice shelves;
    - **Lakes and Rivers**;

    - **Mountains** - mountain ranges and summits;
    - **Physical features** - deserts, plateaus, peninsulas, coasts and other regions.

  Cities and islands have their country in brackets after the name, so that the many cities named Paris can be told apart;

- your own layers, read from files; see Create a Custom Layer;
- **Pins**; see Pins.

With the Amateur Radio functions on, the Layer box also lists the Amateur Radio layers, first; see Prefix Search and Award Tracking.

The layer selected is saved when the program closes.

List

Some layers keep **lists**: named sets of the layer's items that you create. With a list selected, each row has a checkbox, ticked when its item is in the list; tick or clear it to add or remove the item. The lists are saved as soon as they change.

The List box holds:

- **<none>** - no list selected, and no checkboxes;
- the lists of the layer, by name;
- **New List...** - prompts for a name and creates an empty list;
- **Delete List...** - deletes the selected list, once you confirm it.

The layers of the atlas - cities, countries, rivers and the rest - are for looking items up and have no lists, so their List box is disabled. Your own layers and the Pins layer have lists, and so do the Amateur Radio layers, on which the lists track awards. The format of the files that your own layers are read from is described in Create a Custom Layer.

Search

Type a name in the **Search** box and the list shows only the items whose names match. Every word typed must match the start of a word of the name: `new y` finds New York. The ✕ button right of the box clears it.

By default a name matches by its letters, ignoring case. In the **Lists** section of the Settings Dialog, **Name Search** can be set to match names by their sound instead, so that a misspelt name still finds its item: *Gray* finds *Grey*.

The List of Items

The list shows the items' names, and for the layers whose items have a readable id, such as a zone number, a second column with the id. On a layer that holds items of several types, such as **Mountains** or **Seas**, the second column is **Type** instead: Range, Mountain, Gulf, Strait and so on. Click a column header to sort by that column; click it again to reverse the order.

Hover the mouse over a name to see its tooltip.

From a Row to the Map

Double-click a row, or right-click it and pick **Find on Map**, to find its item: the item is highlighted on both maps with a yellow outline, or with a ring at its point, and the map you last used is centred on it and zoomed so that the item spans about half of the map.

Ctrl-click a row, or right-click it and pick **Show Details**, to see its Item Details.

From the Map to a Row

Right-click an item on the map and pick **Find in List**: its row is selected and scrolled into view, its layer is made current if it is not, and the item is highlighted on both maps. The map is not moved.

A click on the map clears the highlight.

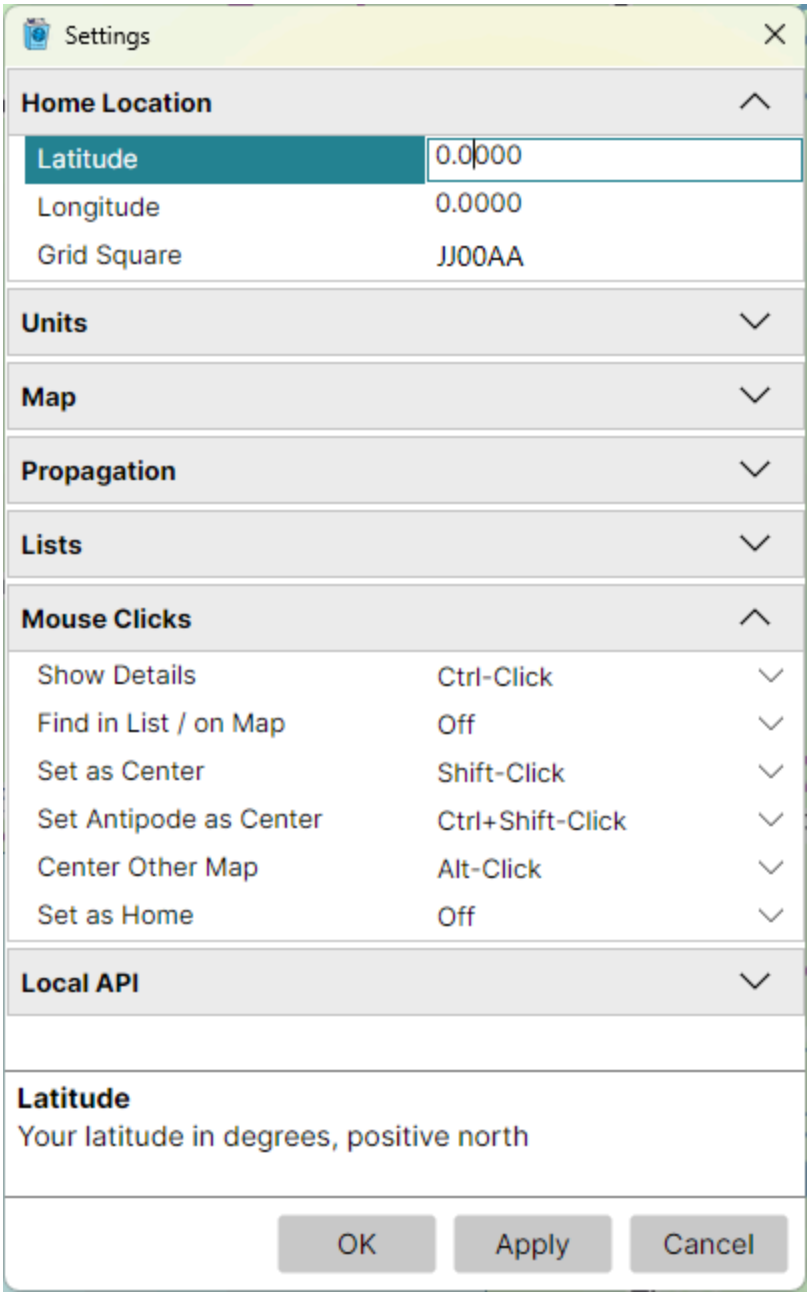
The keys of these clicks can be changed in the **Mouse Clicks** section of the Settings dialog. The **Find in List / on Map** action has no keys by default; give it keys there to find a row's item, or a map item's row, with a single click.

Hiding the Panel

While the Lists panel is hidden, the maps show no highlight, no pins and no award colouring.

Settings Dialog

Tools > Settings opens the Settings dialog: a list of settings in sections, each section folding open to its fields.



- click a field to edit it; its description is shown under the list;
- right-click a field's name and pick **Reset** to return it to its default value;
- **OK** applies the changes and closes the dialog, **Apply** applies them and keeps it open, and **Cancel** discards the changes not yet applied.

A grid square that is not a valid Maidenhead locator, or two mouse click actions given the same keys, are reported when you click OK or Apply, and nothing is

applied until they are corrected.

Home Location

Your location, marked on both maps; see Home Location.

- **Latitude** - in degrees, positive north;
- **Longitude** - in degrees, positive east;
- **Grid Square** - your 4- or 6-character Maidenhead locator. Entering it moves the latitude and longitude to the centre of the square.

Units

- **Distance** - **km** / **m**, **Miles** / **Feet** or **Nautical Miles** / **Feet**: the first unit for the distances in the status bar and on the map, the second for the elevation and the map scale in the status bar.

Map

The settings both maps share:

- **Gray Line Update Interval** - how often the gray line is redrawn as the sun moves, in seconds;
- **Gray Line** - the parts of the gray line drawn while it is on, and where its night shadow begins:
  - **Sun** - the sun glyph at the point where the sun is overhead;
  - **Terminator** - the line between day and night;
  - **Twilight Shadow** - a lighter band between the terminator and the night shadow;
  - **Night Shadow** - the darker shadow over the night side;
  - **Twilight** - where the twilight shadow ends and the night shadow begins: at the end of **Civil**, **Nautical** or **Astronomical** twilight, with the sun 6°, 12° or 18° under the horizon;
- **Value Tooltip** - while an ionospheric map or the propagation map is shown, a tooltip follows the mouse over the map with the map's value under it, as the status bar shows it. Off by default.

Propagation

What the propagation map is computed for; see Propagation Prediction. This section is shown only with the Amateur Radio functions on.

Lists

- **Name Search** - how the Search box of the Lists Panel matches names: **Exact**, by their letters, ignoring case, or **Phonetic**, by their sound, so *Gray* finds *Grey*.

Mouse Clicks

The keys held with a click for each action, on a map or on a row of the Lists panel. Each action can be given **Click**, **Ctrl-Click**, **Shift-Click**, **Alt-Click**, **Ctrl+Shift-Click**, **Ctrl+Alt-Click** or **Alt+Shift-Click**, or **Off** for no keys. No two actions may have the same keys, but any number may be off. Every action is also in the map's context menu, whatever its keys.

Action	Default	What it does
Show Details	Ctrl-Click	shows the Item Details of what is under the click, or of the row clicked
Find in List / on Map	Off	on a map, selects the item clicked in the Lists panel; on a row, finds its item on the map
Set as Center	Shift-Click	centres the azimuthal map on the point clicked
Set Antipode as Center	Ctrl+Shift-Click	centres the azimuthal map on the point opposite the one clicked
Center Other Map	Alt-Click	centres the other map on the point clicked
Set as Home	Off	makes the point clicked your home location, once you confirm it

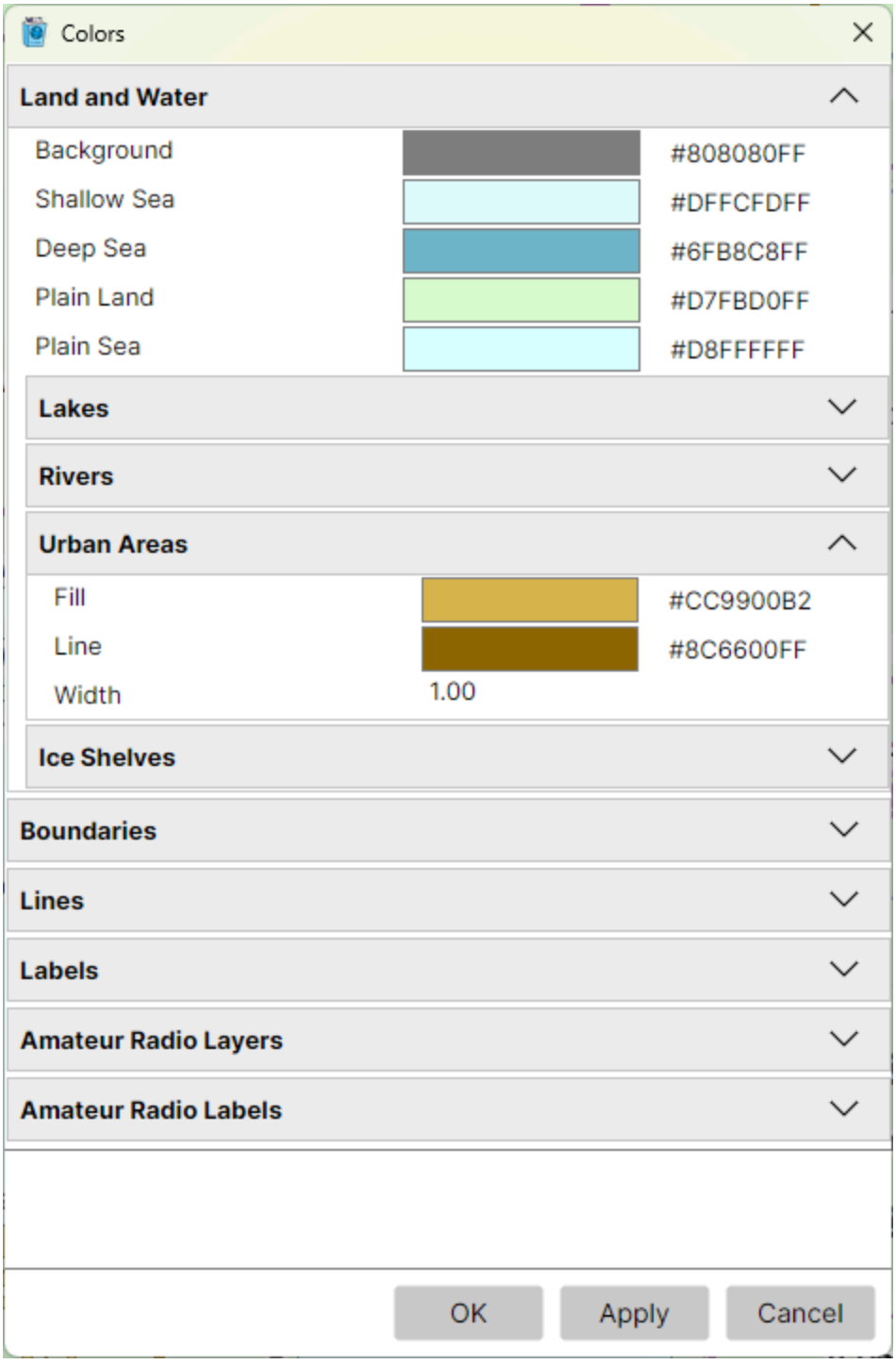
A plain click on a label shows its Item Details unless an action has been given **Click**.

API

The WebSocket API through which other programs on this computer drive the maps: the **Port** they connect to, and the **Allowed Origins**, the web pages that may connect. See the Developer's Guide.

Colors Dialog

Tools > Colors opens the Colors dialog, which sets the colours, line widths and label faces of everything drawn on the map. It works as the Settings Dialog does: sections that fold open, **Reset** on a right-click of a field's name, and **OK**, **Apply** and **Cancel**. **Apply** redraws both maps in the new colours while the dialog stays open, so you can try a colour before you keep it.

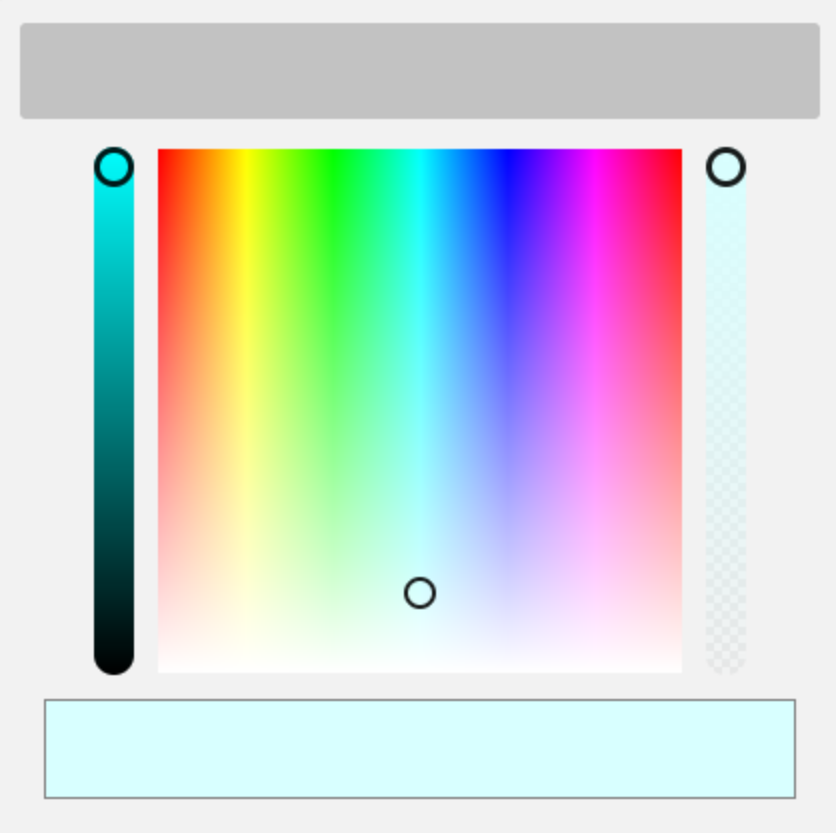


Each colour has an opacity as well, so a fill can let the relief show through it.

Picking a Colour

Each colour field shows a swatch of the colour and its value, `#RRGGBBAA` - red, green, blue and opacity in hexadecimal. Right-click the value and pick **Copy** to copy it to the clipboard.

Click the swatch to open the colour picker:



- **the large square** - the hue from left to right, and the saturation from top to bottom, full colour at the top fading to white at the bottom. Click or drag in it to pick a colour;
- **the slider on the left** - the brightness, from black at the bottom to full brightness at the top;
- **the slider on the right** - the opacity, from fully transparent at the bottom, where the checkerboard shows through, to opaque at the top;
- **the bar at the bottom** - the colour picked so far.

The field takes the new colour as you pick it. Click outside the picker or press **Esc** to close it, then **Apply** or **OK** to redraw the maps in it.

Sections

- **Land and Water** - the background, the shallow and the deep sea, the plain land that replaces the relief when **Land Relief** is off and the plain sea that replaces the ocean depth when **Ocean Relief** is off, the lakes, rivers, urban areas and ice shelves;
- **Boundaries** - the coastline; the borders, which strengthen and widen as the map zooms in; the four fills of the time zones and their boundaries; the date line;
- **Lines** - the graticule, the azimuth and distance lines, the gray line's shadows and terminator, and the highlight of an item found from the Lists Panel;
- **Labels** - each kind of label: countries, big and small cities, island groups, islands, oceans, seas, lakes, rivers, mountain ranges, summits, regions, ice shelves and the graticule's degrees. Each has a colour, a halo colour around the letters, a size, and bold and italic.

With the Amateur Radio functions on, two more sections set the colours of the Amateur Radio layers and labels; see The Amateur Radio Switch.

The colours are saved in a file of their own, `Colors.json`, in the user data folder.

The Amateur Radio Switch

DX Atlas Pro serves two audiences: users who want a world atlas, and Radio Amateurs. All of the Amateur Radio functions are behind one switch, **Tools > Amateur Radio Functions**, so a user who does not need them never meets a Ham Radio term in the menus and lists.

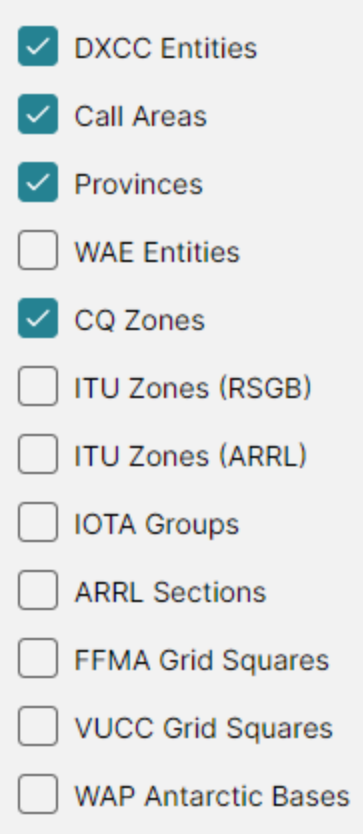
The program asks whether to turn the switch on when it starts for the first time; see Quick Start. Click the menu item to turn it on or off at any time.

What the Switch Shows

With the Amateur Radio functions on, the program adds:

- on each map toolbar, the **Amateur Radio** button, with the Amateur Radio layers, and the **Ionosphere** button, with the ionospheric maps, the propagation map and the aurora;
- on each map toolbar, the **Great Circle** button; see Great Circle;
- in the **Tools** menu, **Ionospheric Parameters...** and **Pre-compute Propagation Maps...**;
- in the **Layer** box of the Lists Panel, the Amateur Radio layers, with their lists for Award Tracking, and the Prefix Search on the DXCC layer;
- in the Item Details, the DXCC entity, call area, province, zones and IOTA groups at the point clicked;
- in the Colors Dialog, the **Amateur Radio Layers** and **Amateur Radio Labels** sections;
- in the Settings Dialog, the **Propagation** section;
- an antenna in place of the house at your home location.

The Amateur Radio Button



The **Amateur Radio** button opens the list of the Amateur Radio layers of its map:

- **DXCC Entities**, **Call Areas**, **Provinces** and **WAE Entities** - see DXCC Entities and Prefixes;
- **CQ Zones**, **ITU Zones (RSGB)** and **ITU Zones (ARRL)** - see CQ and ITU Zones;
- **IOTA Groups** - see IOTA;
- **ARRL Sections** - see ARRL/RAC Sections;
- **FFMA Grid Squares** - see Other Entity Classes;
- **VUCC Grid Squares** - see VUCC Grid Squares;
- **WAP Antarctic Bases** - see Other Entity Classes.

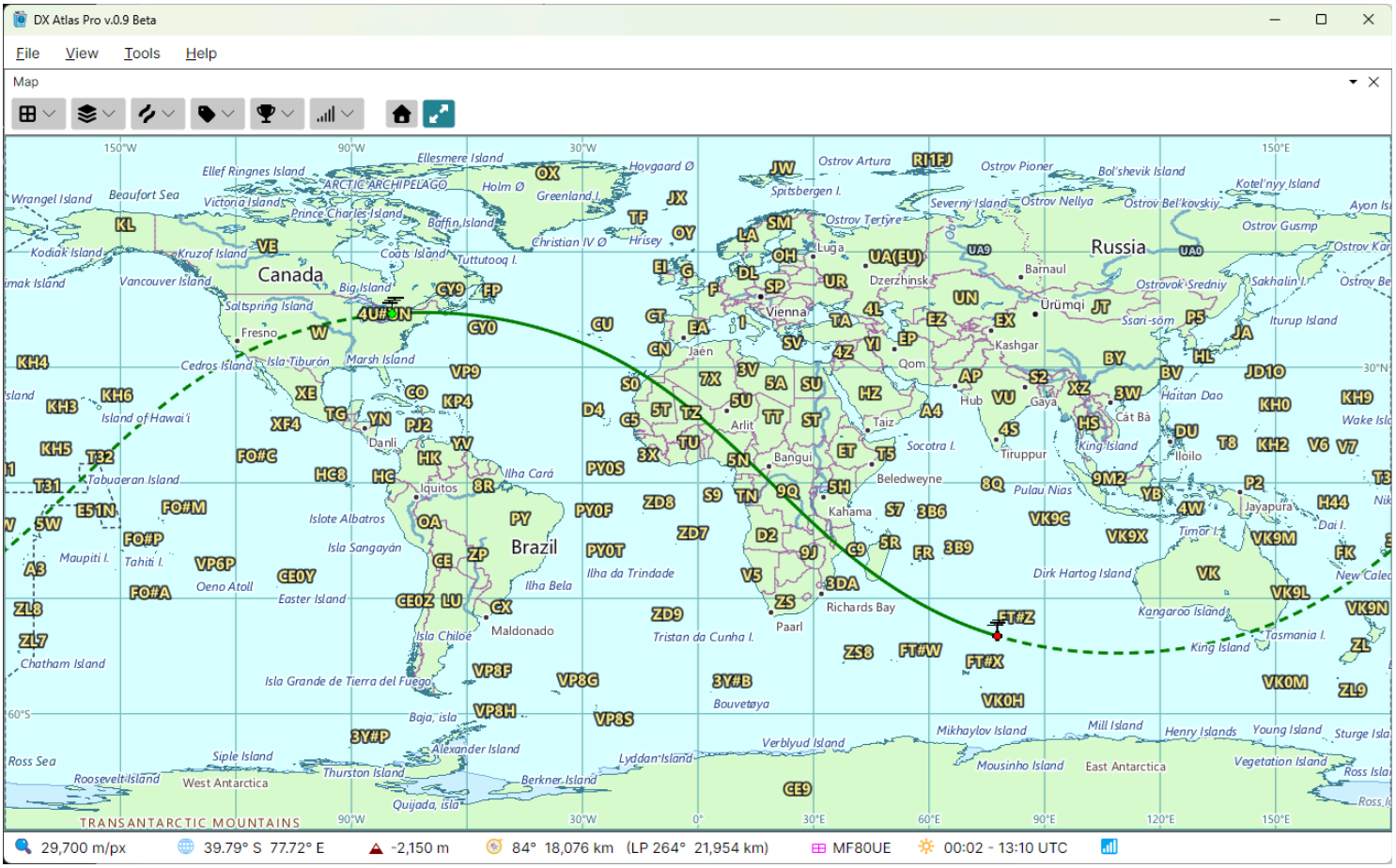
Two Sets of Checkboxes

Each map remembers its checkboxes separately for the two states of the switch. Turning the functions off hides the Amateur Radio layers and keeps the choices you made; turning them on again brings those choices back, and the general layers as you had them in that state. So you can keep, say, a plain atlas with the cities on, and a Ham map with the cities off and the CQ zones on, and switch between them with one click.

A named view saves the state of the switch too.

Great Circle

The **Great Circle** button on the map toolbar, right of the **Home** button, draws the path of a signal from your station. Click it to keep it down: while it is down, the map draws the great circle from your home location to the point under the mouse, the shortest path between them on the globe. Click it again to stop drawing.



To see the path between two other points, press the left mouse button on the first point and drag the mouse to the second: while the button is held, the map draws the great circle, short path and long path, between these points, and the Status Bar shows the bearing and the distance along this path, from the point where you pressed the button. When you release the button, the great circle from your home location returns. While the **Great Circle** button is down, dragging the mouse draws this path instead of moving the map; use the mouse wheel to zoom, or click the button again to drag the map.

The bearing and the distance along the great circle, by the short and the long path, are shown in the Status Bar: the bearing to turn the beam to, and, after **LP**,

the opposite bearing and the distance the other way around the Earth.

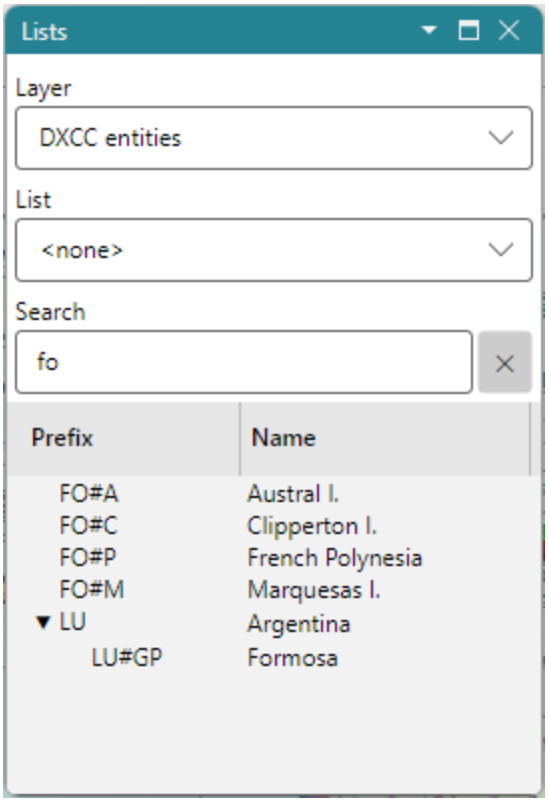
With the Amateur Radio functions on, the home location is marked with an antenna in place of the house.

Prefix Search

The **Search** box of the Lists Panel finds DXCC entities, call areas and provinces. Pick the **DXCC entities** layer, and type any of these:

- a prefix, such as `UA0` or `KH6`;
- a full callsign, such as `VE3NEA` or `KH6/W1AW`;
- the name of an entity or a province, such as `Canada`.

There is no callsign database: the location of a callsign comes from its prefix alone.



- the list shows the DXCC entities whose prefixes match the callsign, and under each one the call area and the province the callsign belongs to, expanded: **VE3NEA** lists Canada, with its call area VE3;
- entities and provinces whose names match the text are listed too, after the prefix matches.

Double-click a row to find the entity on the map, or Ctrl-click it to see its Item Details.

How the Text Is Read

The text is read as a callsign when it is one word of letters and digits, optionally with a `/`, that holds a digit or a `/`, or is one or two letters: `VE3NEA`, `KH6/W1AW`, `UA0K`, `F`. Other text, such as `USA`, is matched only against the names, so it finds the United States and not the entities whose prefixes begin with those letters.

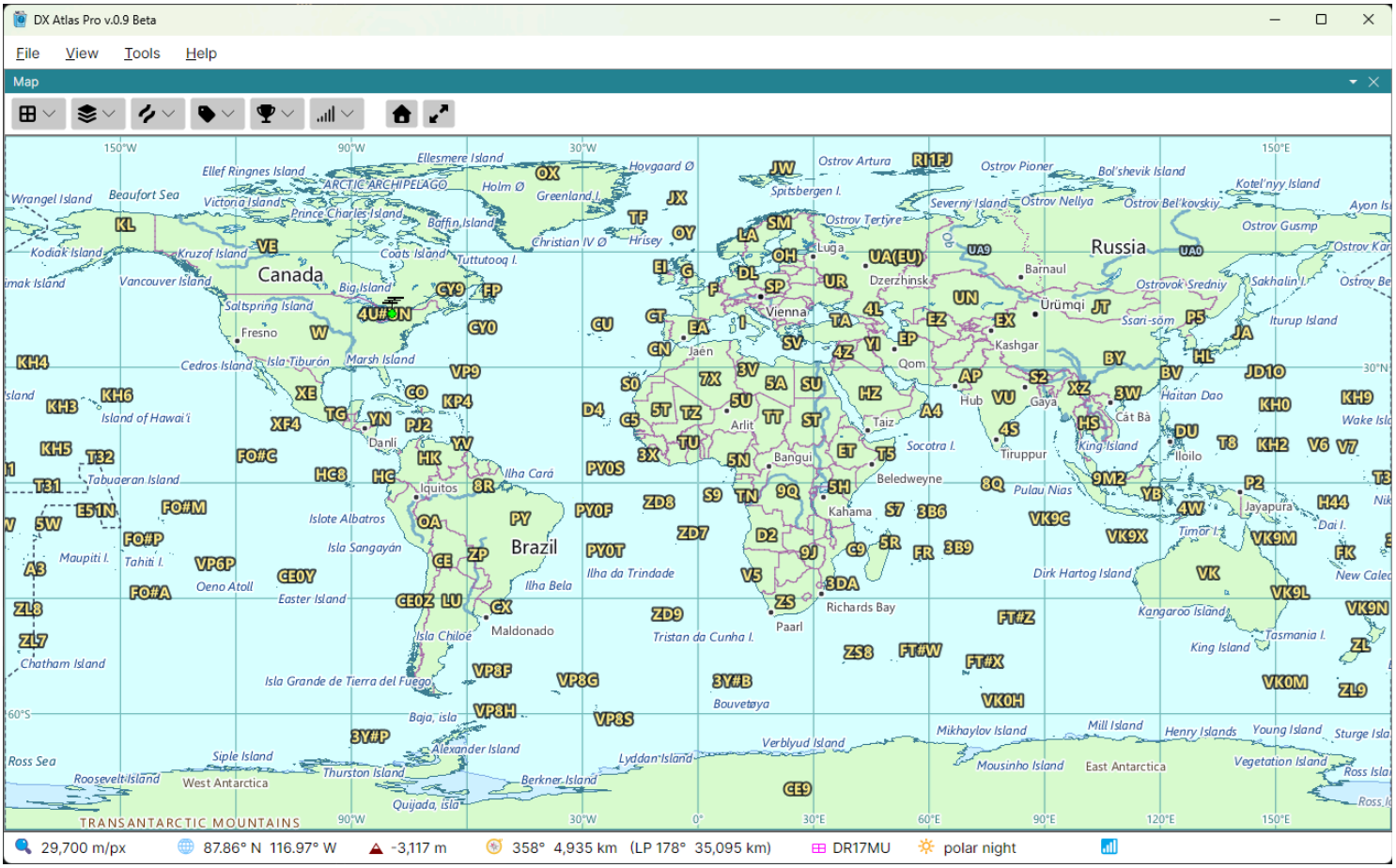
The callsign is matched against the prefix masks of every DXCC entity, call area and province, and every record whose mask matches is listed, not only the most specific one. A portable prefix, such as `KH6/` in `KH6/W1AW`, is matched as a prefix of its own. A partial prefix that no mask matches whole, such as `UT3` or `JT`, lists the entities whose masks begin with it.

The DXCC Tree

On the DXCC layer the list is a tree: each DXCC entity, with its call areas under it, and the provinces or states under them. The prefix column shows each record's prefix, and the triangle before a prefix opens and closes the rows under it.

See also DXCC Entities and Prefixes.

DXCC Entities and Prefixes



On the Map

The first four checkboxes of the **Amateur Radio** button draw the DXCC entities and their subdivisions. Each one switches a set of boundaries and its prefix labels together:

- **DXCC Entities** - the borders of all 340 current DXCC entities, and their prefixes. The borders between countries that are also DXCC entities are drawn by **Political Boundaries** in the Lines menu as well, so the two can be shown separately;
- **Call Areas** - the call areas of the larger entities, such as W4 or VE3, and their prefixes;
- **Provinces** - the provinces, states and oblasts that have prefixes of their own, and their prefixes. Call Areas and Provinces are available only while DXCC Entities is on;
- **WAE Entities** - the countries of the DARC's Worked All Europe list, such as the Shetland Islands (GM/s) or Sicily (IT9), and their prefixes. While it is on,

the WAE borders replace the DXCC, call area and province boundaries, so WAE Entities and DXCC Entities are never on together: ticking one clears the other.

Hover the mouse over a prefix to see the name of its entity, province or call area; for a call area, its definition. Click a prefix to see its Item Details: the prefix, the CQ and ITU zones, the continent, the ADIF number and the local time.

In the Lists Panel

The **DXCC entities** layer of the Lists Panel lists the entities as a tree: each DXCC entity, the call areas under it, and the provinces under them, as the prefixes are structured. The **Prefix** column shows each one's prefix, and the triangle before it opens and closes its rows. Type a callsign in the Search box to find its entity; see Prefix Search.

With a list selected, only the DXCC entities have checkboxes: a DXCC list is a list of entities, and the call areas and provinces under them are there to be looked up. See Award Tracking.

The **WAE countries** layer lists the WAE countries by their prefixes, and the **WAS** layer the 50 US states, Alaska and Hawaii among them, for the Worked All States award.

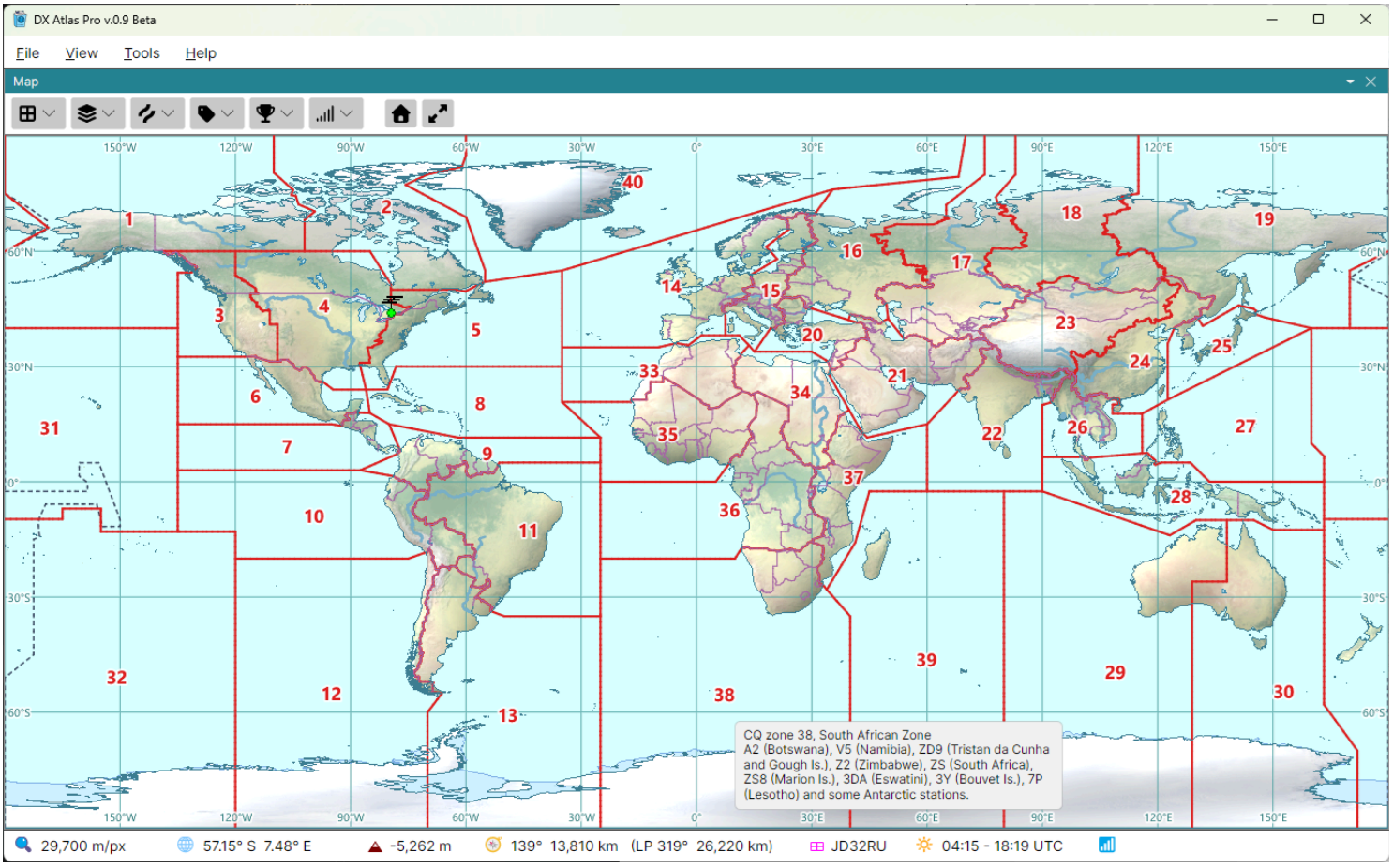
DXCC is a trademark of the ARRL; the DXCC entity list is reprinted with the permission of the ARRL. For the DXCC programme and its rules, see www.arrl.org/dxcc.

CQ and ITU Zones

The **Amateur Radio** button has three zone layers:

- **CQ Zones** - the 40 zones of CQ magazine's WAZ award and the CQ contests;
- **ITU Zones (RSGB)** - the 75 ITU zones as the RSGB defines them for its WITUZ award;
- **ITU Zones (ARRL)** - the 75 ITU zones as the ARRL defines them for the IARU HF Championship.

The two societies draw some ITU zone boundaries differently, so the program has both versions. Each zone set is drawn as filled and outlined areas, each zone labelled with its number. The layers are off by default, but for the CQ zones on the first map.



Tooltips and Details

Hover the mouse over a zone number to see the zone's definition: the prefixes and areas it covers, such as *C3 (Andorra), CT (Portugal), CU (Azores), ...* for CQ zone 14.

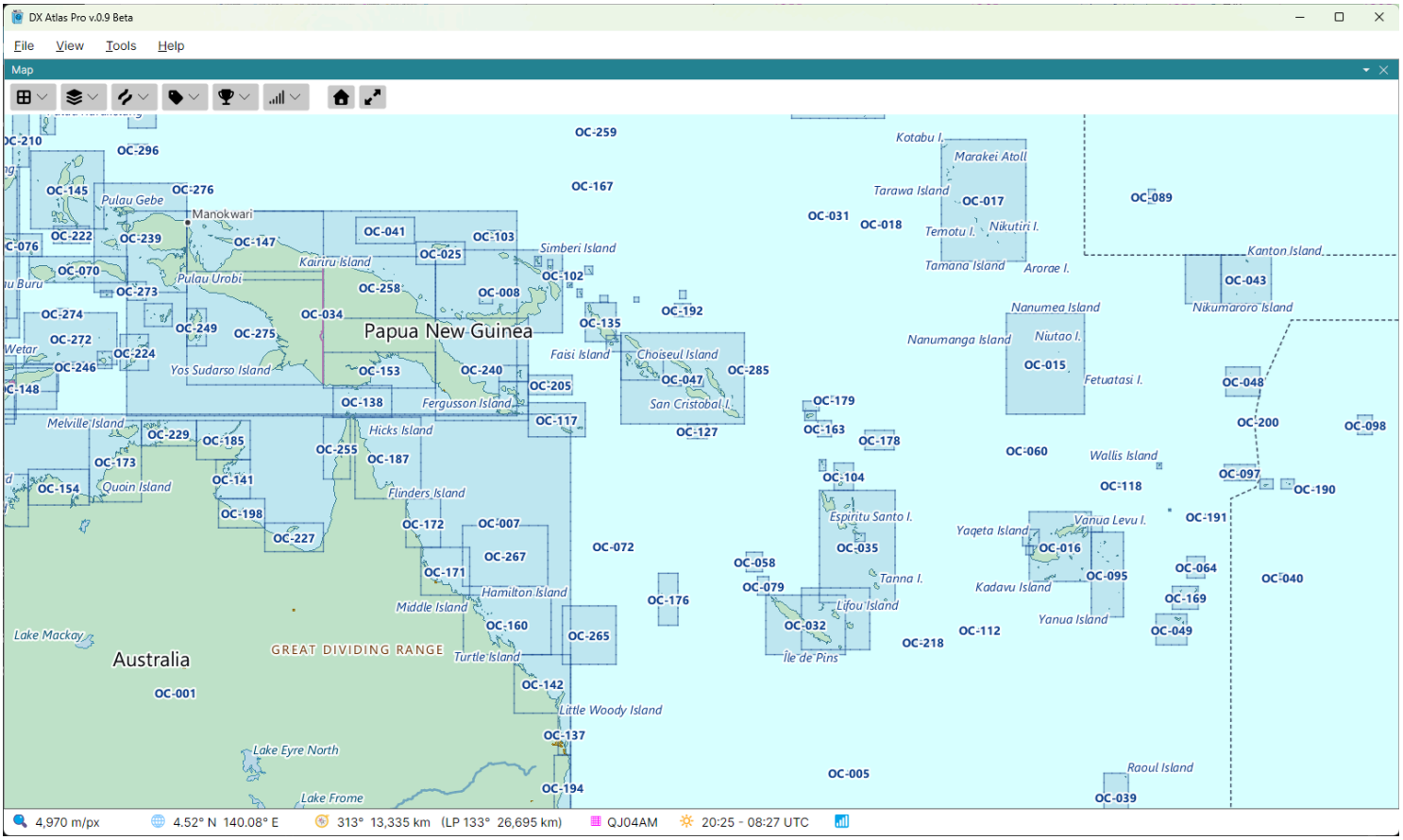
The Item Details of a point always list its CQ zone and its ITU zone, and, where the ARRL's ITU zone differs from the RSGB's, both. A point at sea in no ITU zone is listed as **none**.

In the Lists Panel

The **CQ zones**, **ITU zones (RSGB)** and **ITU zones (ARRL)** layers of the Lists Panel list the zones by number and name, and keep lists of zones worked or confirmed; see Award Tracking.

The colours of the zones and of their numbers are set in the **Amateur Radio Layers** and **Amateur Radio Labels** sections of the Colors Dialog.

IOTA



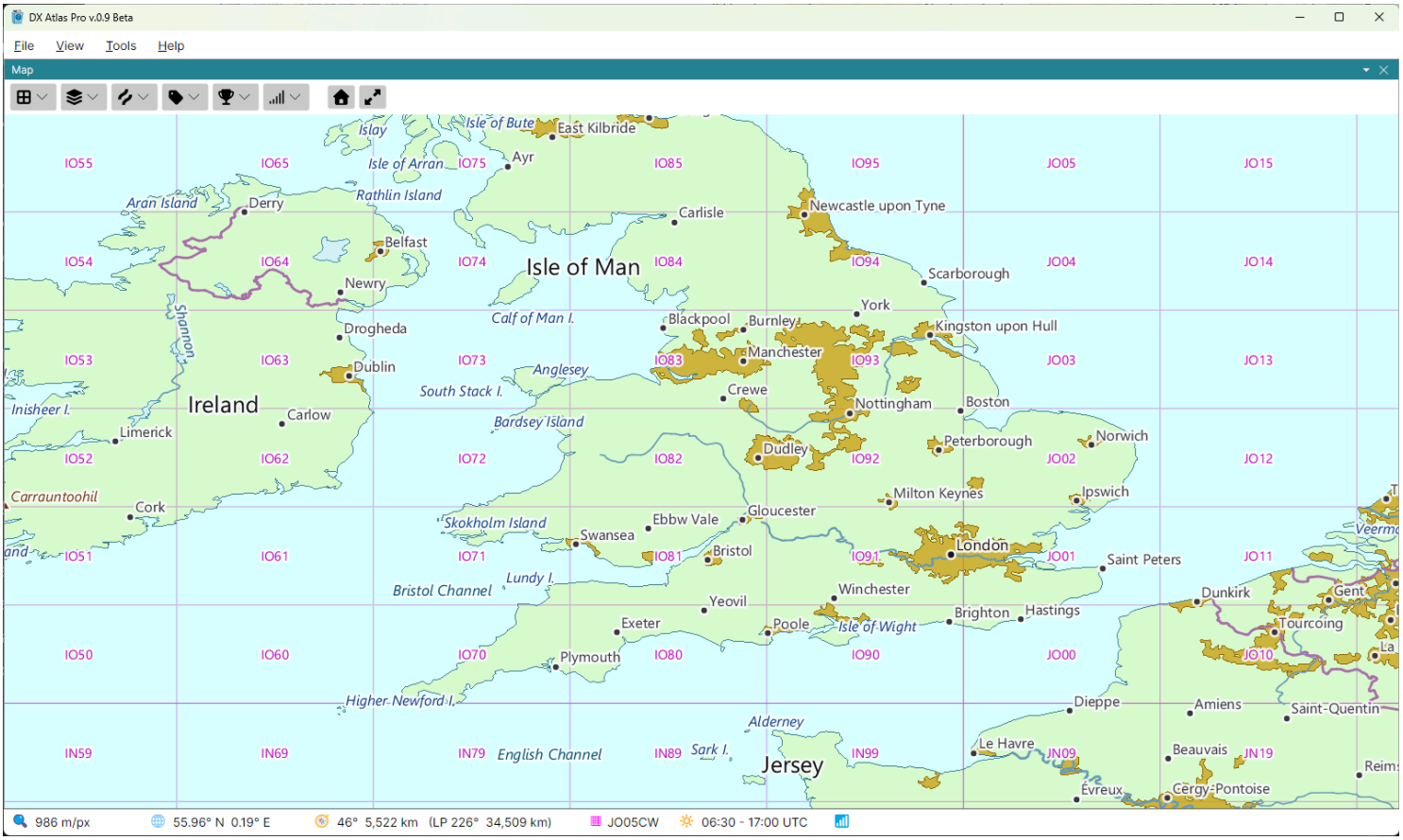
IOTA Groups in the **Amateur Radio** menu draws the 1,178 island groups of the Islands on the Air programme, each as the rectangle that bounds its islands, labelled with its reference, such as AF-001. The layer is off by default.

Hover the mouse over a reference to see the group's name. The rectangles of neighbouring groups overlap, so the Item Details of a point list every IOTA group whose rectangle holds it.

The **IOTA groups** layer of the Lists Panel lists the groups by reference and name, and keeps lists of the groups worked or confirmed; see Award Tracking.

The IOTA groups and their rectangles are used with the permission of the IOTA Committee. For the programme and its rules, see www.rsgbiota.org.

VUCC Grid Squares



VUCC Grid Squares in the **Amateur Radio** menu draws the Maidenhead locator grid, used by the ARRL's VHF/UHF Century Club award and by VHF contesters:

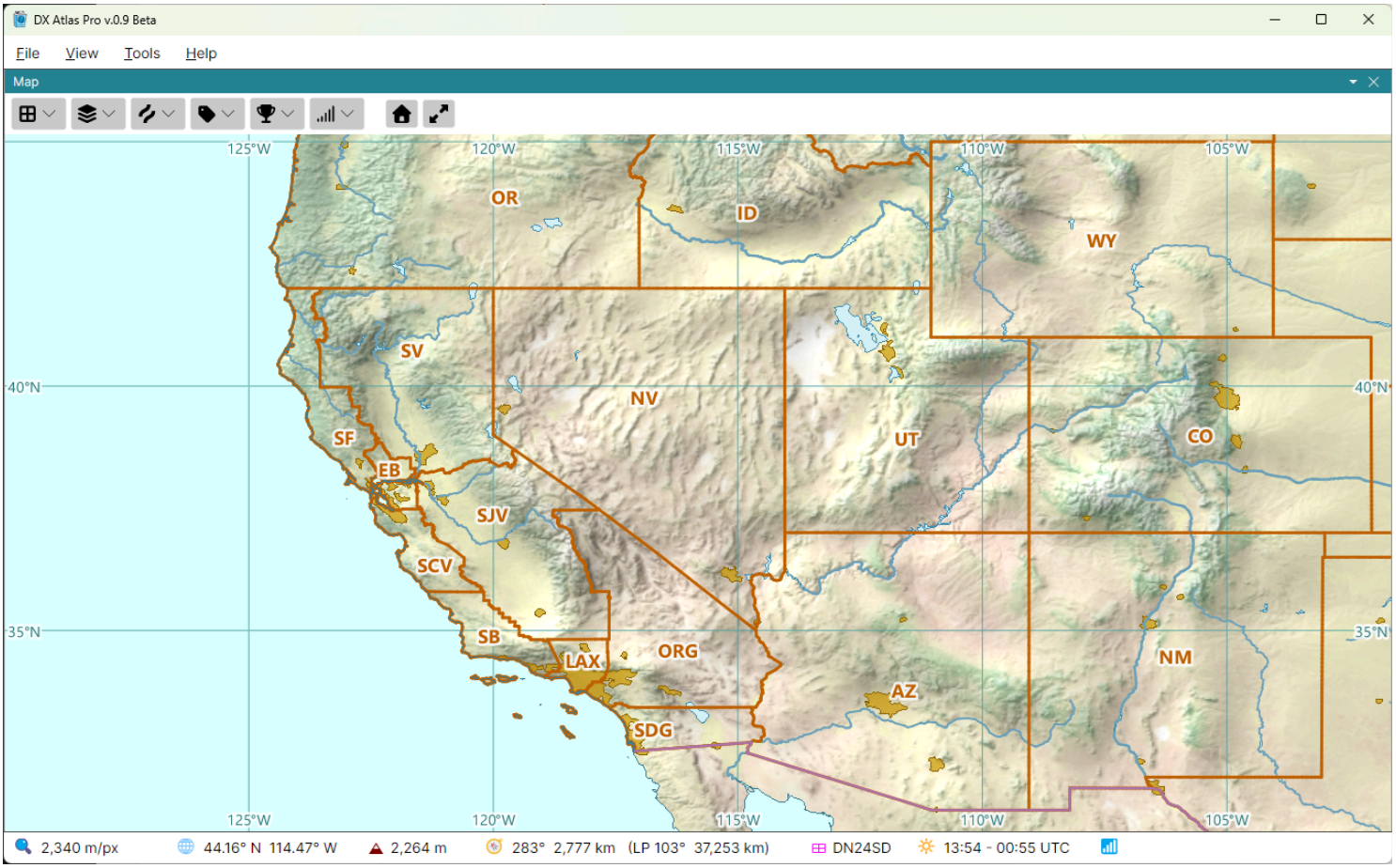
- at a world view, the 324 fields, 20 by 10 degrees, labelled **AA** to **RR**;
- zoomed in, the 32,400 squares, 2 by 1 degrees, labelled with their four characters, such as **FN03**.

The grid is drawn by the program as you zoom, so it is sharp at every scale.

The status bar always shows the 6-character locator of the point under the mouse; see Status Bar.

ARRL/RAC Sections

ARRL Sections in the **Amateur Radio** menu draws the 85 sections of the ARRL and of Radio Amateurs of Canada, used in the ARRL Sweepstakes, Field Day and other contests, each labelled with its code, such as **EMA** or **ONS**. The layer is off by default.



Hover the mouse over a code to see the section's name. The Item Details of a section list its call area.

The **ARRL/RAC sections** layer of the Lists Panel lists the sections by code and name, and keeps lists of the sections worked; see Award Tracking.

Other Entity Classes

FFMA Grid Squares

FFMA Grid Squares in the **Amateur Radio** menu draws the 488 grid squares of the ARRL's Fred Fish Memorial Award, the squares of the 48 contiguous United States, each labelled with its locator, such as **CM79**. The layer is off by default.

The **FFMA grid squares** layer of the Lists Panel lists the squares, and allows you to keep lists of the squares worked and confirmed on 6 metres; see Award Tracking.

WAP Antarctic Bases

WAP Antarctic Bases in the **Amateur Radio** menu shows the Antarctic bases and sites of the Worldwide Antarctic Program, each as a dot labelled with its WAP reference, such as **ARG-01**. The layer is off by default.

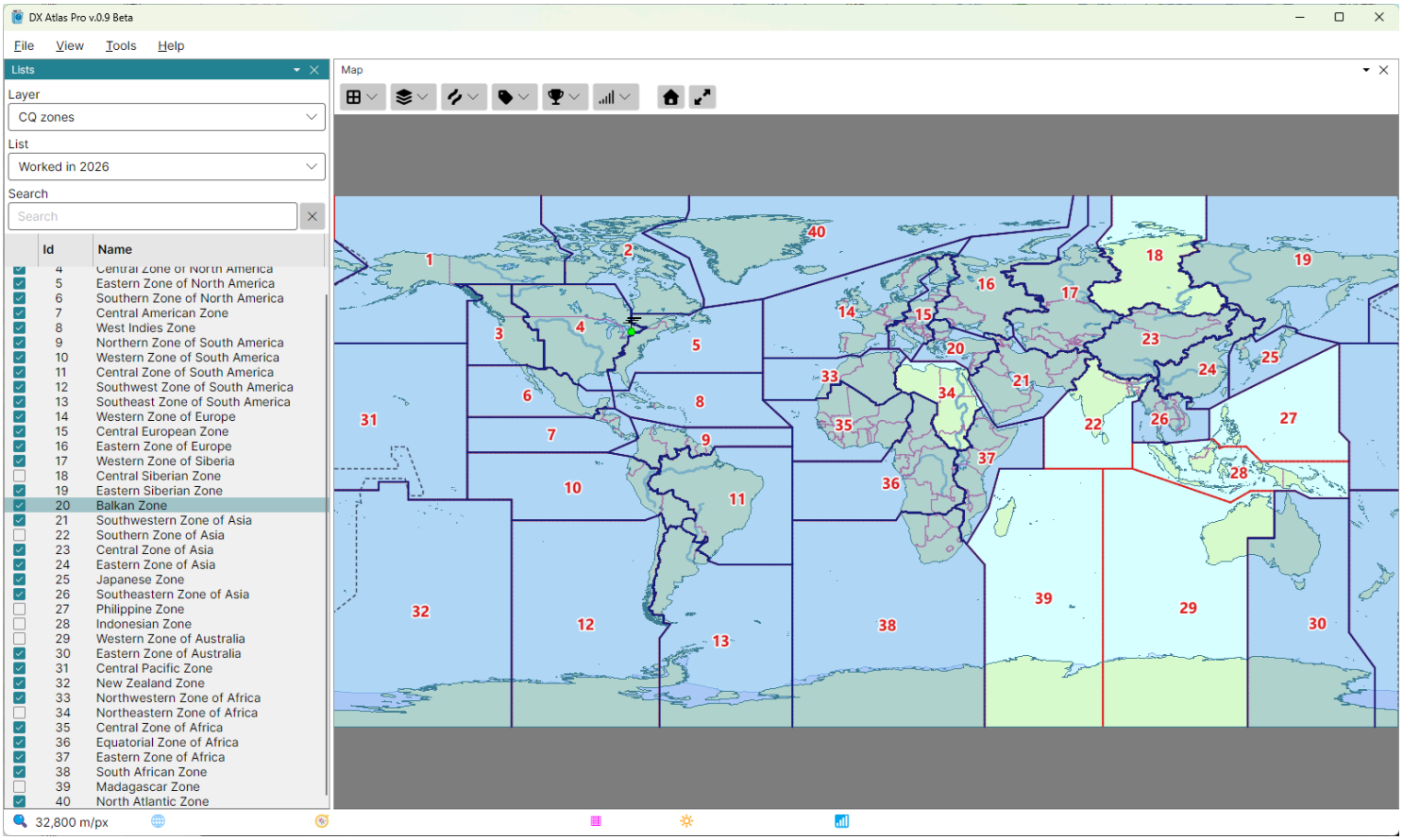
Hover the mouse over a reference to see the base's name and its location.

The **WAP Antarctic bases** layer of the Lists Panel lists the WAP references and keeps lists of the references worked or confirmed; see Award Tracking.

The WAP base and station list is used with the permission of Massimo Balsamo IK1GPG, of the Worldwide Antarctic Program.

Award Tracking

The lists of the Lists Panel track your progress towards awards: a list is a named set of the items of one layer - DXCC entities, CQ zones, IOTA groups - that you tick as you work or confirm them, and the map paints the items of the list selected.



Award Layers

These layers keep lists:

- **DXCC entities**, **WAE countries** and **WAS**;
- **CQ zones**, **ITU zones (RSGB)** and **ITU zones (ARRL)**;
- **IOTA groups**, **ARRL/RAC sections** and **FFMA grid squares**;
- **WAP Antarctic bases**;
- your own layers; see Create a Custom Layer.

Creating Lists

Create one list for each record you keep. A list holds one flag per item, so worked and confirmed are separate lists, as are bands and modes: "Worked",

"Confirmed", "160 m", "CW", or "160 m confirmed".

1. pick the layer in the **Layer** box;
2. in the **List** box, pick **New List...** and type the list's name;
3. tick the checkbox of each item you have; type in the **Search** box to find an item quickly. On the DXCC layer, typing the callsign of a station you worked finds its entity; see Prefix Search.

The lists are saved as soon as a checkbox changes, in the file `Lists.json` in the user data folder, with a backup copy, `Lists.json.bak`, beside it. **Help > Data Folder...** opens that folder.

Award Colouring

While a list is selected, both maps paint its items over the regular map:

- an item with an area - an entity, a zone, an IOTA group's rectangle - is filled semi-transparent and outlined;
- an area too small to see at the current zoom, such as the Vatican, is drawn as a circle too, so it still shows at a world view.

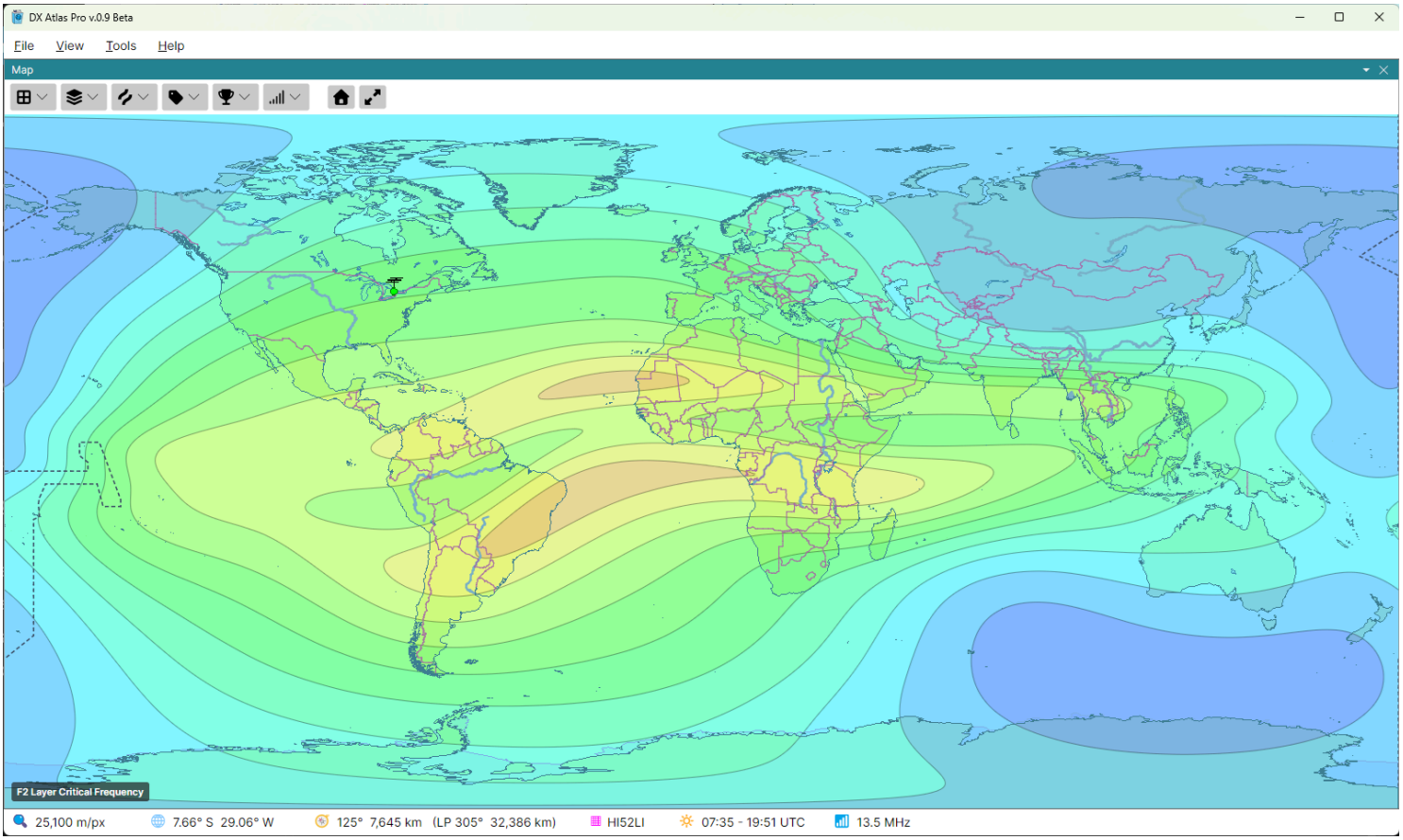
Ticking or clearing a checkbox paints or clears its item at once. With **<none>** selected, or while the Lists panel is closed, nothing is painted. The items of your own layers are not painted.

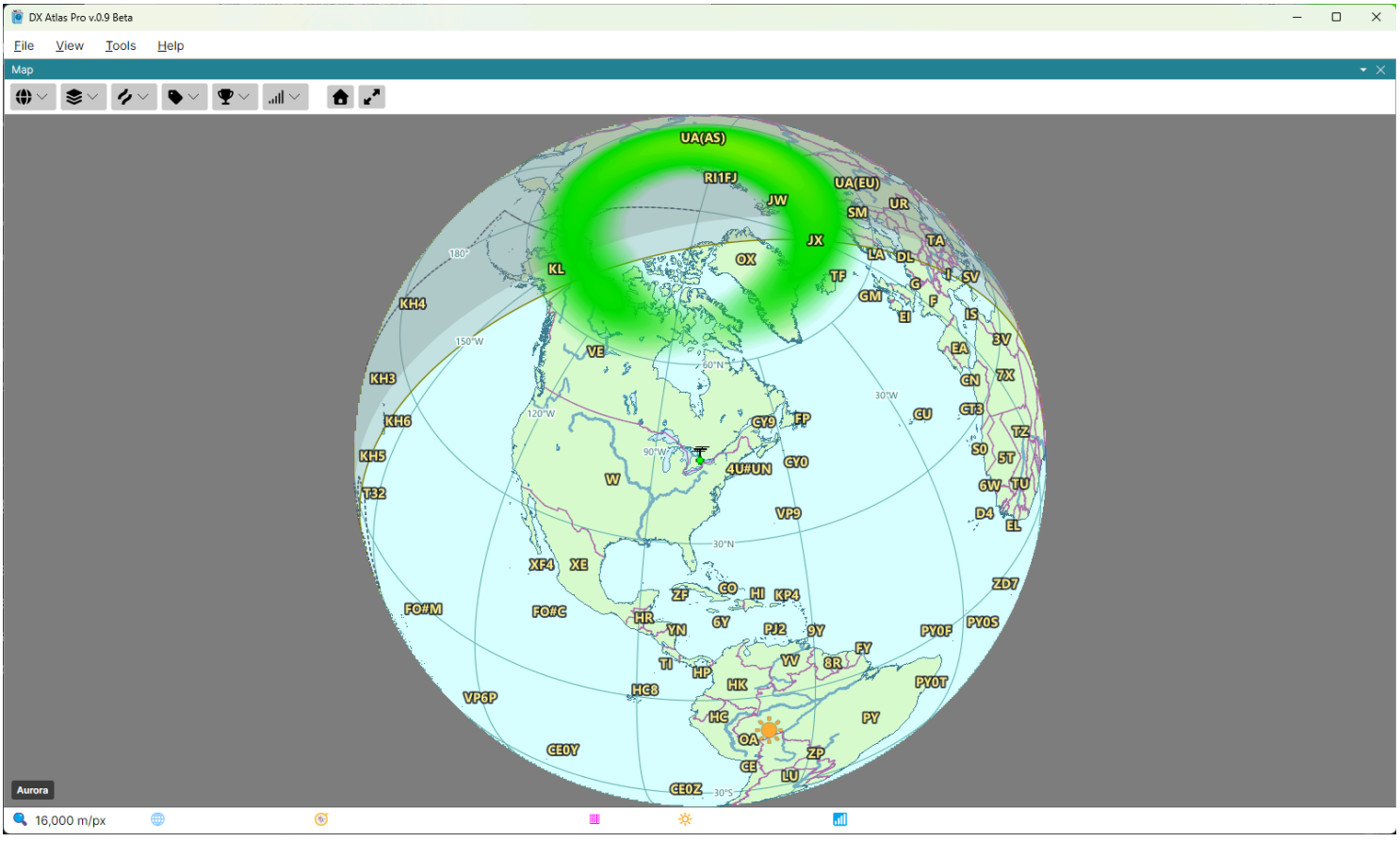
The fill and the outline colours are **Award Coloring** in the **Amateur Radio Layers** section of the Colors Dialog.

Deleting a List

Pick **Delete List...** in the List box and confirm it. The list and its ticks are deleted.

Ionospheric Maps





The **Ionosphere** button on the map toolbar, shown with the Amateur Radio functions on, draws maps of the ionosphere over the world map. The maps are computed by the program with the IRI-2026 model of the International Reference Ionosphere, for the moment the Clock panel shows: the current time, or a fixed time in the past or the future. What each map tells a DXer is explained in Reading the Maps for DX'ing.

The Ionosphere Button



- **HF Propagation**, **Band** and **Long Path** - the propagation map; see Propagation Prediction;
- **None**, or one of the ionospheric maps:
  - **F2 Layer Critical Frequency** - foF2, the highest frequency the F2 layer reflects straight up, in MHz; see foF2 and MUF(3000);
  - **MUF for 3000 km** - MUF(3000), the highest frequency that one 3000 km hop through the F2 layer supports, in MHz, banded at the amateur bands; see foF2 and MUF(3000);
  - **F2 Layer Peak Height** - hmF2, the height of the F2 layer's peak, in km; see hmF2;
  - **E Layer Critical Frequency** - foE, in MHz; see foE;
  - **D Layer Peak Electron Density** - NmD, in units of 10^8 electrons per cubic metre; see NmD;
- **Magnetic Dip** - the dip equator and the lines of 20° and 40° magnetic dip on either side of it, with dashed lines 2000 km beyond the 40° lines; see

Magnetic Dip;

- **Aurora** - the auroral oval, the probability of visible aurora, from the NOAA Space Weather Prediction Center's OVATION model; see The Aurora Oval.

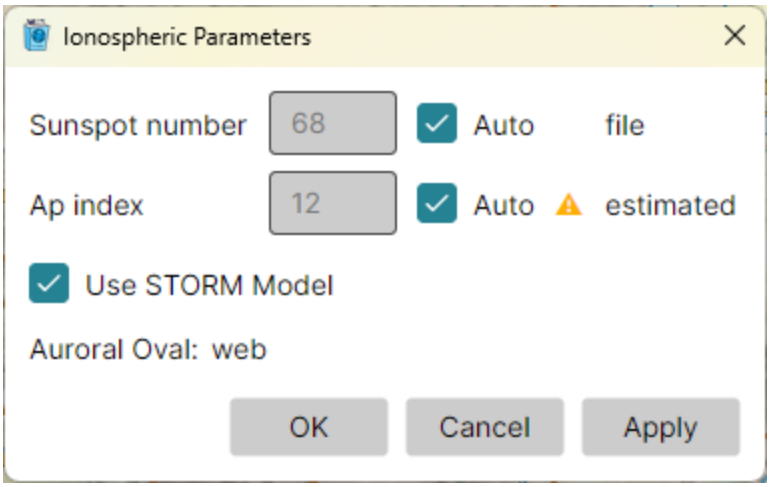
The maps between the separators exclude one another, and exclude HF Propagation too: picking one replaces the map shown. **Magnetic Dip** and **Aurora** are checkboxes of their own, drawn alone or over any map. Each map panel has its own choice, so the two maps can show different maps.

Reading a Map

Each map is drawn in bands of colour, with the legend in the lower left corner of the map: its name and its unit. The status bar's 📶 cell shows the value at the point under the mouse - `7.2 MHz`, `312 km`, and with Magnetic Dip and Aurora on, `dip 23°` and `aurora 15%` after it. While a map is being computed, the cell shows ….
To see the value beside the mouse instead of looking down at the status bar, turn on **Value Tooltip** in the **Map** section of **Tools > Settings**: a tooltip with the same value then follows the mouse over the map.

The Ionospheric Parameters Dialog

Tools > Ionospheric Parameters... shows the solar and geomagnetic indices the maps are computed from, and where each one came from:



- **Sunspot number** - the smoothed sunspot number of the map's month;
- **Ap index** - the geomagnetic activity index;
- **Use STORM Model** - corrects foF2 for geomagnetic storms, from the recent history of the Ap index, on the ionospheric and the propagation maps. It is disabled while there is no Ap data;

- **Auroral Oval** - the state of the aurora data.

With **Auto** ticked, a box shows the value the maps use. Clear **Auto** to enter a value of your own, then click **Apply** to see the maps with it - what the bands would look like at a sunspot number of 200, say. The values entered last until the program closes; on the next start, both rows return to Auto.

Where the Values Come From

After each value, one word says where it came from; hover over it for the date:

- **file** - the index file installed with the program, until a download arrives;
- **web** - downloaded;
- **user** - entered by you;
- **estimated** - the map's date is past the end of the data, so the last value is used: the sunspot number at any date, the Ap index only within 9 hours of the last value;
- **not fetched** - the aurora, while no map shows it; it is downloaded only then;
- **no data** - nothing to use. With no Ap data the storm model is skipped; with no aurora data no aurora is drawn.

An amber ⚠ before the word means a download failed and older data is used; hover over it for the error and the time of the next attempt. The ⟳ button after the word retries the download at once. A red ✕ means there is no data at all.

The Downloads

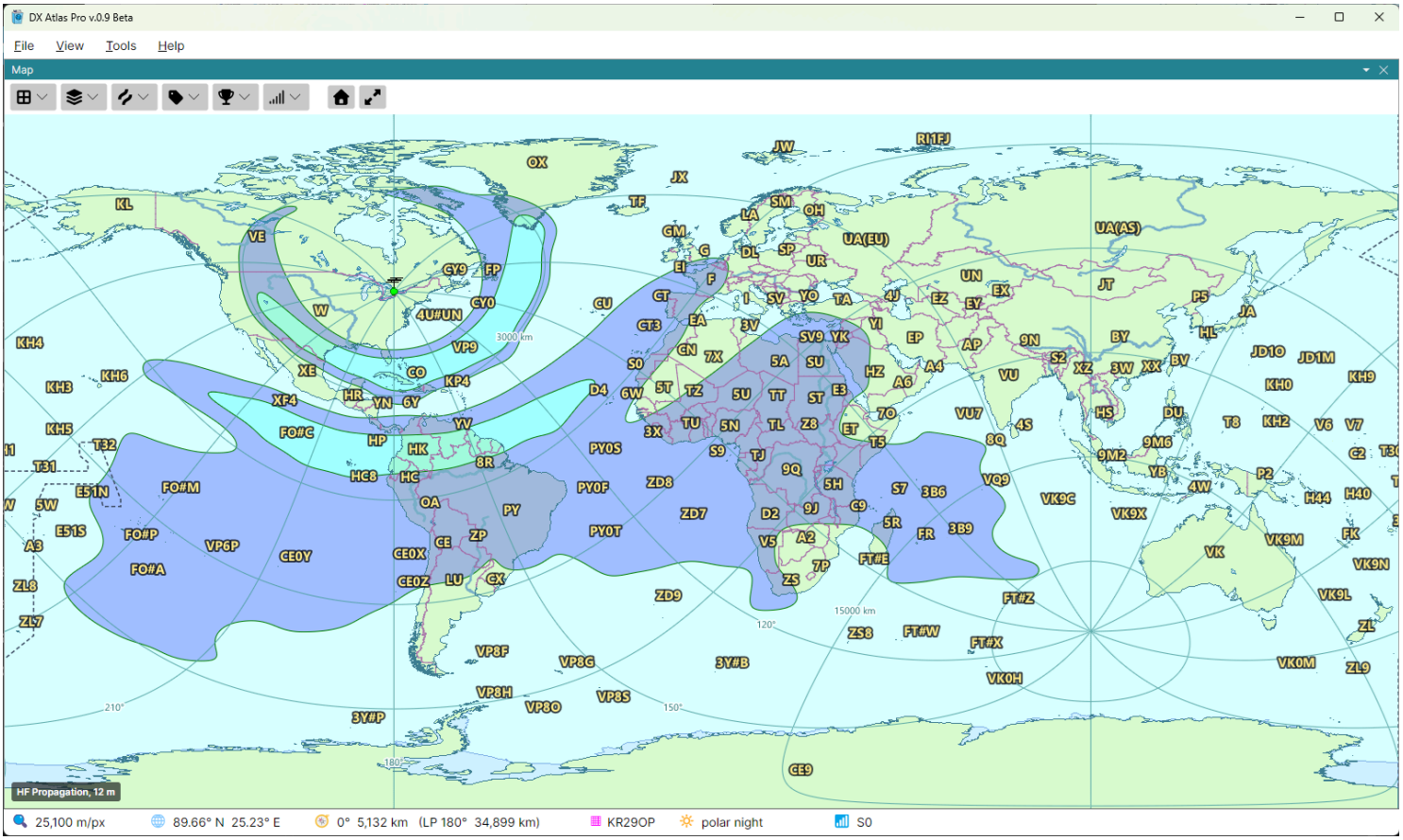
The program downloads the indices by itself, and only while a map that uses them is shown:

- the Ap index from the GFZ Helmholtz Centre for Geosciences, every 3 hours, as each new value is published;
- the sunspot numbers from irimodel.org, once a month;
- the aurora from the NOAA Space Weather Prediction Center, every 15 minutes while a map shows it.

Without an Internet connection, the maps are computed from the index files installed with the program, and from the last downloaded data.

The real-time aurora, the Ap index and the sunspot numbers are available only as long as the NOAA Space Weather Prediction Center, the GFZ Helmholtz Centre for Geosciences and irimodel.org provide them on the web. If a source stops publishing its data or changes its format, the maps that use it fall back to the data already downloaded or installed with the program.

Propagation Prediction



The propagation map shows how strong your signal would be at every point of the globe on one band, at the time the Clock panel shows. It is computed by the program with the DVOACAP engine based on the VOACAP algorithms, from your home location, the band, and your station as set in the **Propagation** section of the Settings Dialog.

Showing the Map

Open the **Ionosphere** button on the map toolbar:

- tick **HF Propagation** to show the map in place of an ionospheric map;
- pick the band in the **Band** submenu: 80, 60, 40, 30, 20, 17, 15, 12 or 10 m;
- tick **Long Path** to compute the signal along the long path, the longer arc of the great circle around the Earth, instead of the short path.

The band and Long Path are the same for both maps; HF Propagation is ticked for each map separately.

Reading the Map

The map is drawn in bands of S-meter readings, from S1 up to S9+12 dB, with a green line at each band edge; below S1 it is left clear. The legend in the lower left corner names the band, with **LP** after it on the long path. The status bar's 📶 cell shows the reading at the point under the mouse: `S7`, or `S9 + 10 dB`; with **Value Tooltip** on in the **Map** section of **Tools > Settings**, a tooltip with the same reading follows the mouse over the map.

How to read the map, and how many S units to add or subtract when your station differs from the settings, is explained in HF Propagation on the *Reading the Maps for DX'ing* page.

Live Updates

The map is computed for a quarter hour of UTC, and changes when the clock enters the next one. With the clock live, the program computes the maps of the next 3 hours in the background, so they are ready when they are needed. With **Use STORM Model** ticked in the Ionospheric Parameters dialog, the map follows geomagnetic disturbances as the Ap index is downloaded; see Ionospheric Maps.

A change of the band, the path, the home location or a setting starts the map again at once; while it is computed, the status bar shows …. The maps already computed are kept on disk, so a return to earlier settings, or a restart, needs no new computation.

The Propagation Settings

The **Propagation** section of the Settings dialog sets what the map is computed for:

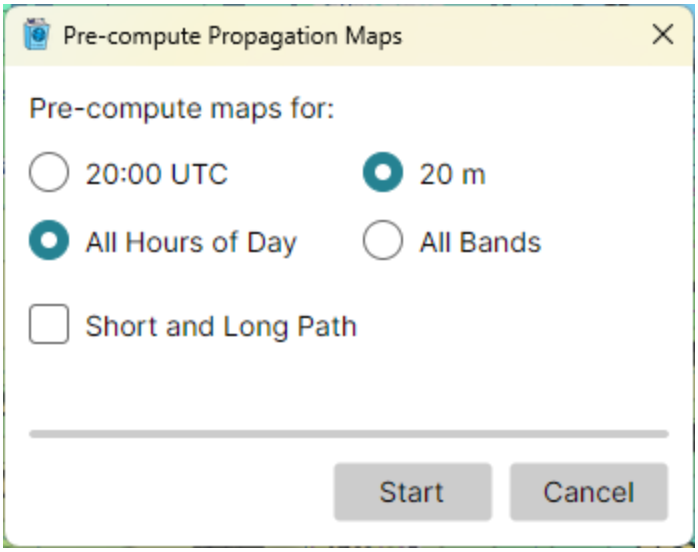
- **Transmitter Power, W** - your transmitter's power, 100 W by default;
- **Transmit Antenna Gain, dBi** - the gain of your antenna over an isotropic one, the same in every direction;
- **Receive Antenna Gain, dBi** - the gain of the antennas at the receiving end;
- **Minimum Takeoff Angle, °** - the lowest elevation angle the antennas radiate at, 3° by default;
- **Man-made Noise** - the noise at the receiving end: **Industrial**, **Residential**, **Rural** or **Remote**;
- **Grid Resolution** - the spacing of the points the map is computed at, **3°** or **1.5°**. The finer grid takes four times as long;

- **Threads** - the number of processor threads that compute the map, 0 for one per processor.

To compute the maps of a whole day in advance, see Pre-computing Propagation.

Pre-computing Propagation

A propagation map takes a second or so to compute. To step through the hours of a day, or through the bands, without waiting for each map, compute them in advance: pick **Tools > Pre-compute Propagation Maps...**, shown with the Amateur Radio functions on.



The Dialog

Under **Pre-compute maps for:**, choose:

- the current quarter hour, shown as its UTC time, or **All Hours of Day** - the 96 quarter hours of the 24 hours from now;
- the band the map shows, or **All Bands**;
- **Short and Long Path** - ticked, both paths are computed; cleared, the path the map shows.

Click **Start**. The dialog counts the maps done and shows a progress bar; maps already computed are counted as done at once. While it runs, **Start** becomes **Stop**.

When the batch is done or stopped, the dialog says so and stays open, so you can start another; its **Cancel** button becomes **Close**. Closing the dialog stops a batch that runs.

When the Maps Are Used

The maps are kept on disk, in the user data folder, for the settings they were computed with. They are shown at once when the Clock panel steps to their time, or when you pick their band or path.

A change of the home location, of a propagation setting, or of the sunspot number or Ap index stops the batch, and the maps already computed no longer match; the dialog reports it. All Hours of Day at All Bands on both paths is 1,728 maps, which takes several minutes even on a fast computer.

Reading the Maps for DX'ing

A propagation prediction answers one question: on this band, at this hour, how strong will the signal be along the great circle between two stations? It answers it for the median day of the month, and on any given day the answer may be wrong. The path may be absorbed, the band may close early, or the signal may arrive from a direction no prediction offered.

The ionospheric maps of DX Atlas Pro are a road atlas for those days. They show the state of each layer of the ionosphere across the whole globe, so you can see why a predicted path failed, and find another route for the signal: a skewed path, a chordal hop, the long path, or a band that is open along a different arc.

This page explains what each map tells a DXer. The controls that show the maps are described in Ionospheric Maps and Propagation Prediction.

The ionospheric maps are computed with the IRI-2026 model, which gives the monthly median ionosphere for the sunspot number of the month. With **Use STORM Model** on, foF2 is corrected for the geomagnetic activity of the last hours, which brings the maps closer to the ionosphere of the day during a storm.

foF2 and MUF(3000)

foF2, the critical frequency of the F2 layer, is the highest frequency that the layer reflects straight up. **MUF(3000)**, the maximum usable frequency of a 3000 km hop, is about three times foF2. A path is open when the band is below MUF(3000) at its control points, about 1500 km from each end of the path; a shorter hop has a lower MUF. The distance circles of the **Azimuth/Distance** lines (Layers, Lines and Labels) are drawn every 3000 km, the length of an F2 hop, so each circle marks where a hop from their centre lands.

Where foF2 rises steeply along the path, the signal may travel in **chordal hops**, without touching the ground between, and stay strong. Where the great circle crosses a region of low foF2, the signal may arrive by a **skewed path** through higher foF2 off the great circle, from an unexpected bearing.

hmF2

hmF2 is the height of the F2 layer's peak, typically 250 to 400 km. The higher the layer, the longer each hop, the fewer hops a long path needs, and the smaller the factor between foF2 and MUF(3000).

foE

foE, the critical frequency of the E layer, follows the sun: up to about 4 MHz at noon in summer, almost nothing at night. On the low bands in daytime, the E layer can keep a low-angle signal from reaching the F2 layer, reflecting it in hops of at most 2000 km or absorbing it. The map shows the regular E layer only, not sporadic E.

NmD

NmD is the electron density of the D layer, which absorbs HF signals on the day side, the more the lower the frequency. On 160, 80 and 40 metres it decides whether a daytime signal arrives at all; the low bands open where the D layer fades, near the gray line and at night.

Gray Line

Near the terminator, the D layer has faded while the F2 layer is still ionized, so absorption on the low bands is low and the signals are strong. Propagation on 160, 80 and 40 metres is enhanced when both stations are near the terminator: both at sunrise, both at sunset, or one at sunrise and the other at sunset.

At the height of the F2 layer, about 300 km, the sun rises earlier and sets later than on the ground: the layer stays sunlit until the sun is about 17° below the ground's horizon, an hour or more after sunset on the ground, and comes into sunlight as long before sunrise.

Astronomical twilight, which ends with the sun 18° under the horizon, corresponds to sunrise and sunset at the F2 layer's height. Set **Twilight** to **Astronomical** in the **Map** section of the Settings Dialog: the night shadow then begins where the F2 layer goes dark, and the lighter band between the terminator and the night shadow is where the F2 layer is still sunlit while the ground is dark.

The terminator's path across the globe changes with the season, so a gray-line opening to a given station happens only in some months. To find them, show the

Gray Line (Layers, Lines and Labels), click the sunrise or sunset lock button in the Clock panel, and step the date by one month. The time stays at your sunrise or sunset; the months in which the gray line passes over the DX station are the months when both stations are at the gray line.

Magnetic Dip

The **magnetic dip** is the angle of the Earth's magnetic field to the horizontal. At the dip equator the field is horizontal, and there the ionization of the F2 layer is lifted and moved north and south by the combined action of the electric and the magnetic fields. This **equatorial ionization anomaly** leaves two crests of high foF2 on either side of the dip equator, which the foF2 map shows on the day side, strongest in the afternoon and the evening.

The crests support **trans-equatorial propagation** (TEP) on 10 and 6 metres: between stations on opposite sides of the dip equator, roughly the same distance from it, the signal travels from crest to crest in a chordal hop, at frequencies far above the MUF of ordinary hops. The lines of 40° dip on either side bound the anomaly, and the dashed lines 2000 km beyond them mark how far from the anomaly a station can be and still reach it.

The Aurora Oval

The **aurora oval** is the ring around each magnetic pole where particles from the Sun enter the upper atmosphere. It grows and moves towards the equator during geomagnetic storms. HF signals that cross the oval are absorbed, and those that get through arrive weak and fluttering: a polar path, such as Europe to the west coast of North America, is the first to suffer.

The oval also opens the high bands. **Auroral E**, the ionization of the E layer in the oval, reflects and scatters signals on 10, 6 and 2 metres; stations beam towards the oval rather than towards each other, and the signals have a characteristic rasp.

The map shows the probability of aurora from the NOAA Space Weather Prediction Center's OVATION model, downloaded every 15 minutes while the map is shown.

HF Propagation

The **HF Propagation** map shows the signal strength from your home location, on the band and at the hour set, at every point of the globe, computed with the VOACAP model for the median day of the month, and with **Use STORM Model** on, corrected for the geomagnetic activity of the day. The reading is in S units of 6 dB, for the transmitter power and the antenna gains set in the **Propagation** section of the Settings Dialog.

When a station differs from the settings, shift the reading:

- every 6 dB of power or antenna gain is one S unit: four times the power, or an antenna of 6 dBi in place of 0 dBi, adds one S unit, and a quarter of the power takes one away;
- the gains at both ends count, so a beam at the receiving station adds as much as one at yours;
- doubling the power adds only 3 dB, half an S unit.

Finding an Alternate Route

When the predicted path fails, read the maps together:

- **a polar path absorbed in the aurora oval** - show the Aurora over the HF Propagation map. If the short path crosses the oval, try the long path (**Long Path** in the Ionosphere menu), or a skewed path that passes south of the oval, where the foF2 map shows the band is supported;
- **a path through low foF2** - show the foF2 or MUF(3000) map, and look along and beside the great circle for a region where the band is below the MUF. A signal skewed through it arrives off the great-circle bearing, so turn the beam towards that region;
- **a path across the dip equator on 10 or 6 metres** - show the Magnetic Dip lines over the foF2 map. In the afternoon and evening, the equatorial crests may carry the signal in chordal hops well above the predicted MUF;
- **a low band closed in daytime** - show the NmD map and the Gray Line. Absorption falls first where the D layer fades along the terminator, so the low bands open along the gray line.

Fix the Clock and step it through the hours to see how each map, and so each route, changes through the day.

Further Reading

- Carl Luetzelschwab, K9LA, k9la.us;
- OH3AC, *F-Region Propagation and the Equatorial Ionospheric Anomaly*, QEX, November 2016;
- Dean Straw, N6BV, Visalia 2010 paper on reading predictions in S units;
- the VOACAP papers.

Create a Custom Layer

You can add layers of your own to the Lists Panel: mountain summits you climbed, lighthouses, the parks of a park award. Each layer is a JSON file in the `Layers` folder of the user data folder, and appears in the **Layer** box under its own name. A user layer has lists, as the award layers do, so you can track which of its items you have visited or worked.

The Layers Folder

Help > Data Folder... opens the user data folder, which holds the settings, the lists and the caches:

- on Windows, `%APPDATA%\Afreet\Products\DxAtlasPro`;
- on Linux, `~/.config/Afreet/Products/DxAtlasPro`.

Create a folder named `Layers` there, and put the layer files in it. The files are read when the program starts, so restart it after adding or editing a file.

The File Format

One file is one layer, and the file name, without `.json`, is the layer's id: rename a file and its lists are lost. An example:

```
{
  "name": "Summits",
  "idColumn": true,
  "entities": [
    { "id": "W7A/AE-001", "name": "Humphreys Peak", "lat": 35.346, "lon":
-111.678 },
    { "id": "W7A/AE-002", "name": "Agassiz Peak", "lat": 35.326,
"lon": -111.677,
      "tooltip": "3766 m" }
  ]
}
```

- **`name`** - the layer's name in the Layer box;
- **`idColumn`** - `true` to show the items' ids in a column of their own beside the names, `false` to show the names alone;

- **`entities`** - the items.

Each item has:

- **`id`** - unique in the layer; the lists keep the items by their ids;
- **`name`** - shown in the list and searched; the id when it is missing;
- **`lat`** and **`lon`** - the item's position, in degrees, positive north and east;
- **`tooltip`** - optional, the text shown when the mouse is over the name;
- **`geometry`** - optional, the item's area, as a GeoJSON `Polygon` or `MultiPolygon`.

Reference Items

Instead of its own position, an item can name an item that the program already has, with `ref` written as `class:id`. It then takes the position, the area and the tooltip of that item, and an `id` or `name` given beside the `ref` replaces that item's own:

```
{ "id": "AL", "ref": "province:W.W4.W4(AL)" }
```

The program does not show the ids of its own items, so reference items are mainly for layers built from the program's data. The layer of the US states that ships with the program is written this way, and is a working example to copy: on Windows, `C:\Program Files\Afreet\DxAtlasPro\assets\entities\layers\WAS.json`, and on Linux, `assets/entities/layers/WAS.json` in the program folder.

How a User Layer Is Shown

A user layer is not drawn on the map as a whole. When you find one of its items from the list, the item is highlighted: by its area, by the area of the item its `ref` names, or by a ring at its point.

Large layers are fine: a layer of 180,000 summits is searched as fast as a small one.

Errors

A file that cannot be read does not stop the program. When it starts, a message lists what could not be read - a file that is not valid JSON, a `ref` that names no item, an id used twice in one layer - and the rest of the file is loaded.